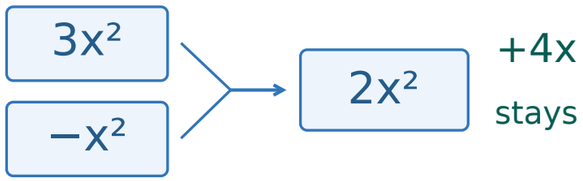
Algebra 1

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 1.1

Collect like terms

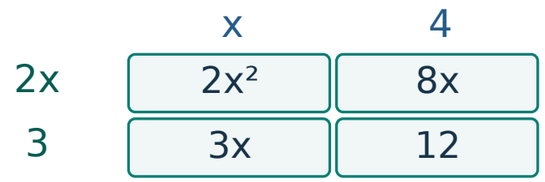


$ax^n + bx^n = (a + b)x^n$

Remember: Only combine matching variables AND powers.

02 / 1.2

Multiply every term by every term



$(2x+3)(x+4) \rightarrow 2x^2+11x+12$

$a(b + c) = ab + ac$

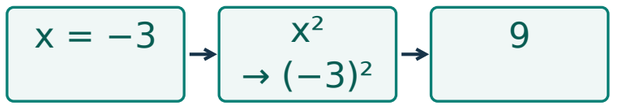
$x^m x^n = x^{m+n}$

m terms × n terms → mn products.

Remember: Index rule here uses non-negative integer powers.

03 / 1.3

Substitute a signed value



Bracket the negative value before squaring.

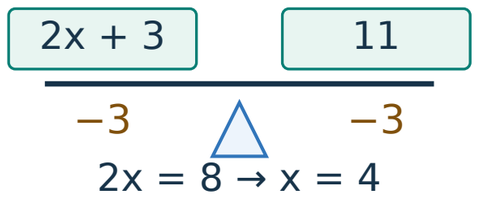
Replace each variable by its whole signed value in brackets

Denominator ≠ 0.

Remember: $(-3)^2 = 9$, but $-3^2 = -9$.

04 / 1.4

Keep an equation balanced



Do the same valid operation to BOTH sides

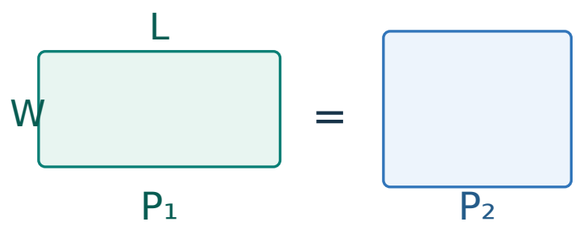
clear every denominator

Never divide by zero.

Remember: A multiplier applies to every term on both sides.

05 / 1.5

Match two perimeters



Rectangle P = 2L + 2W

Set $P_1 = P_2$ when the perimeters are equal.

Remember: Equal boundary lengths do not imply equal areas.

06 / 1.5

Subtract a removed area



Unshaded area = whole area − shaded area.

Remember: Use square units; a minus before brackets changes every sign.

Algebra 1

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

07 / 1.6

Divide every term

$$(6x^2 + 4x) \div 2x$$

$6x^2 \div 2x$ → $3x$ + $4x \div 2x$ → 2

$(A + B - C)/d = A/d + B/d - C/d$

$x^m/x^n = x^{m-n}$.

Remember: Require $d \neq 0$ and $x \neq 0$ for the index division rule.

08 / 1.6

Reverse a multiplication grid

	x	3
x	x^2	$3x$
2	$2x$	6

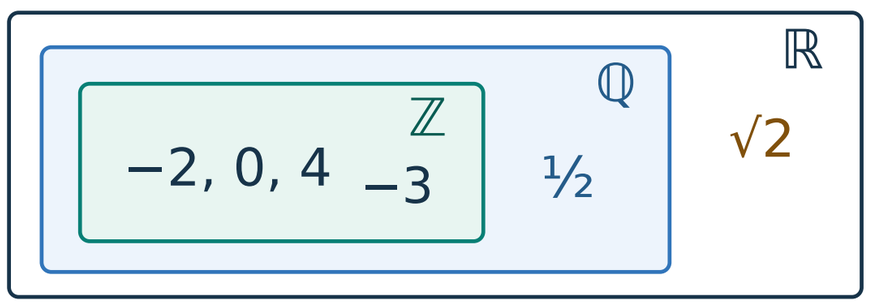
$$x^2+5x+6 = (x+2)(x+3)$$

Dividend = divisor × quotient + remainder.

Remember: Divisor ≠ 0; leave spaces for missing powers.

09 / 1.7

Locate a number family



$\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$

rational numbers are p/q with integers p, q and $q \neq 0$.

Remember: Recurring decimals are rational; √2 is irrational.

10 / 1.8

Solve and shade an inequality

$$-2x \geq -6 \rightarrow x \leq 3$$



Filled endpoint: 3 is included.

Multiply or divide by a negative → reverse <, >, ≤ or ≥.

Remember: Adding or subtracting a negative does not reverse the sign.

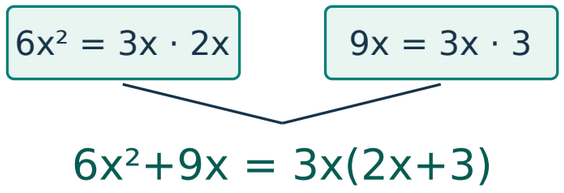
Read the labels. Diagrams illustrate relationships and may not be to scale. Full workings and narration are in the interactive guide.

Factors

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 2.1

Take out the highest common factor

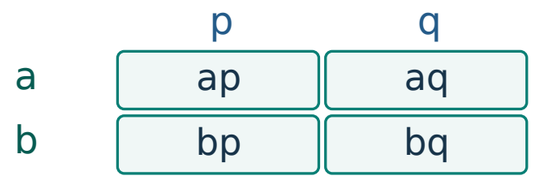


HCF = numerical HCF × lowest power common to every term.

Remember: Check the remaining bracket for further factors.

02 / 2.2

Group to make matching brackets



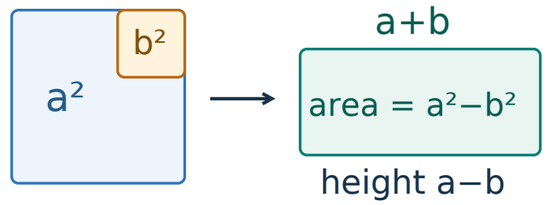
Both rows contain the same bracket: (p+q).

ap + aq + bp + bq = (a + b)(p + q).

Remember: Taking out a negative factor reverses signs inside its bracket.

03 / 2.3

Recognise a difference of squares

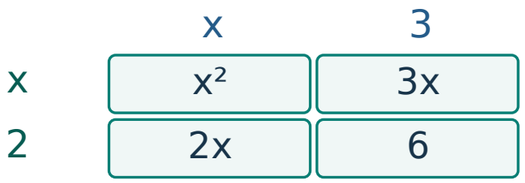


a² − b² = (a − b)(a + b).

Remember: Two squares separated by subtraction; not a²+b² over the reals.

04 / 2.4

Find two numbers for a monic quadratic



p=2, q=3: sum 5; product 6.

x² + bx + c = (x + p)(x + q): p + q = b

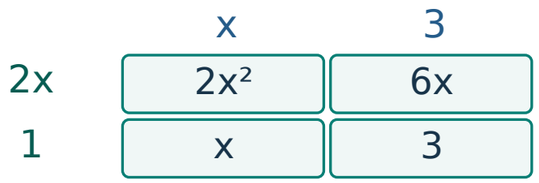
pq = c

x² ± 2ax + a² = (x ± a)².

Remember: Check BOTH the sum and the product.

05 / 2.4

Split the middle term when a ≠ 1



2x²+7x+3: split 7x into 6x+x.

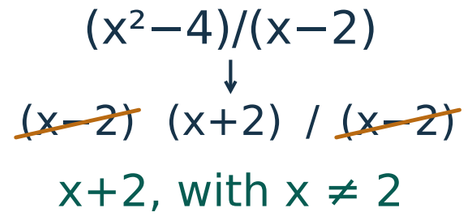
ax² + bx + c: find m + n = b and mn = ac

split bx, then group.

Remember: Use ac for the product, rather than c alone.

06 / 2.5

Cancel factors and keep restrictions



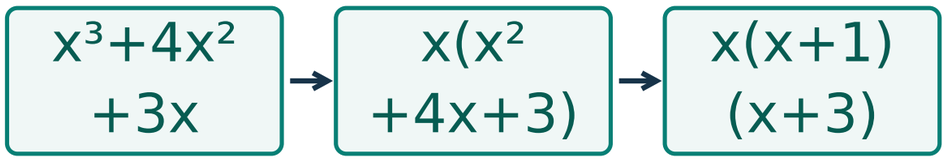
N/D: factor first, cancel common factors only where original D ≠ 0.

Remember: A cancelled denominator factor still creates an excluded input.

Factors

07 / 2.5

Factor a cubic before cancelling



First extract x; then factor the quadratic.

Use HCF or grouping → linear factor × quadratic → further factors.

Remember: Cancel complete factors, while retaining original exclusions.

Sets

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 3.1

Read membership, subset and size



B ⊆ A; 2 ∈ A; n(A)=4

a∈A: member

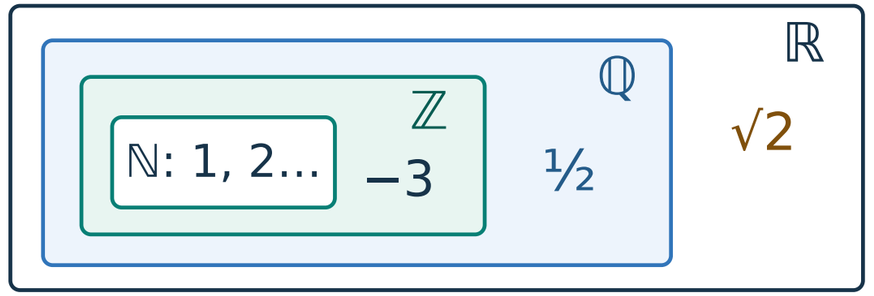
B ⊆ A: every B member is in A

n(A): number of distinct members.

Remember: {0} has one member; ∅ has none.

02 / 3.1

Place numbers in nested families



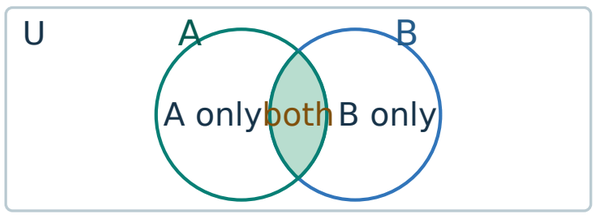
ℕ ⊂ ℤ ⊂ ℚ ⊂ ℝ

irrational = real but not rational.

Remember: Use the convention given for whether ℕ includes 0.

03 / 3.1

Choose union, intersection or complement



∪: either

∩: both

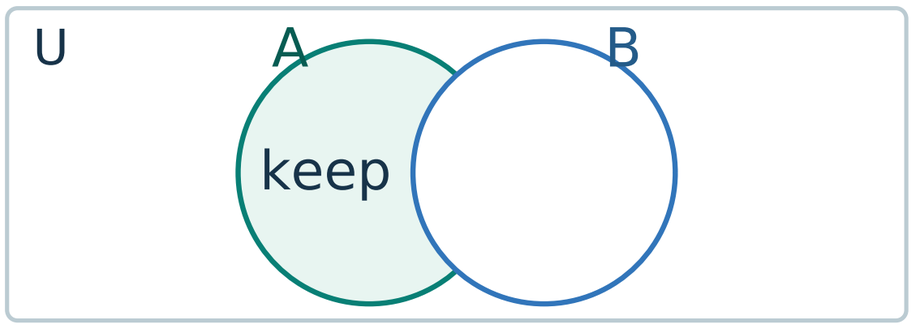
A′ = U \A

n(A′) = n(U) − n(A). ∪ and ∩ are commutative and associative.

Remember: Union includes overlap. The complement depends on U.

04 / 3.2

Find A but not B



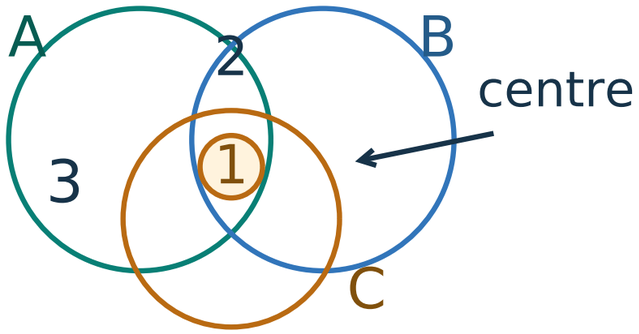
A \B = A ∩ B′.

Remember: Difference is generally neither commutative nor associative.

Sets

05 / 3.3

Fill a three-set diagram from the centre



AB-only = n(A ∩ B) − n(A ∩ B ∩ C)

n(A ∪ B ∪ C)

= n(A) + n(B) + n(C)

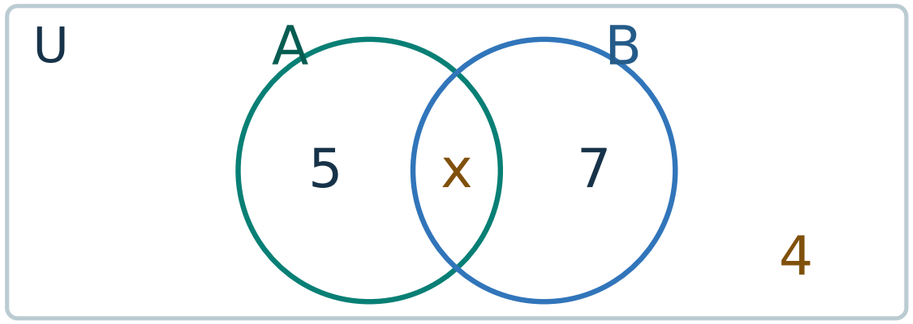
− n(A ∩ B) − n(A ∩ C) − n(B ∩ C)

+ n(A ∩ B ∩ C).

Remember: Pair intersections include the centre unless “only” is specified.

06 / 3.4

Find a missing region or total



n(U)=20: 5+x+7+4=20

Disjoint regions add once

Union = n(U) − outside.

Remember: Counts of people must be non-negative integers.

Sets

07 / 3.4

Find the smallest or largest overlap

N=20, a=12, b=11



overlap can range from 3 to 11

max(0, a + b − N) ≤ overlap ≤ min(a, b)

union = a + b − overlap.

Remember: a and b are set sizes inside a universe of size N.

08 / 3.4

Distinguish people from memberships



3 people → 1+2+3 = 6 memberships

Union = $E_1 + E_2 + E_3$

Membership total = $E_1 + 2E_2 + 3E_3$.

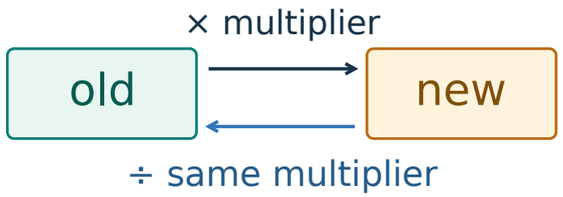
Remember: E_k counts people in exactly k sets; apply ratios to named regions.

Applied Arithmetic

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 4.1

Move forwards or backwards through a percentage



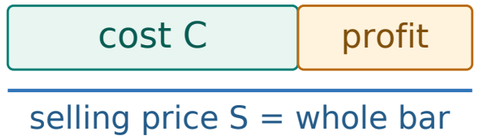
New = old(1 ± r/100)

old = new/(1 ± r/100).

Remember: Reverse using the same non-zero multiplier.

02 / 4.1

Choose the correct profit percentage base



Profit = S − C

markup% = 100 × profit/C

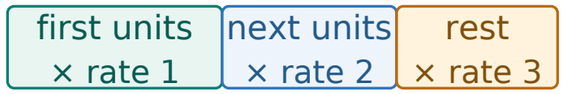
margin% = 100 × profit/S

loss% = 100 × loss/C.

Remember: C=cost, S=selling price. Markup uses C; margin uses S.

03 / 4.2

Charge each usage band separately



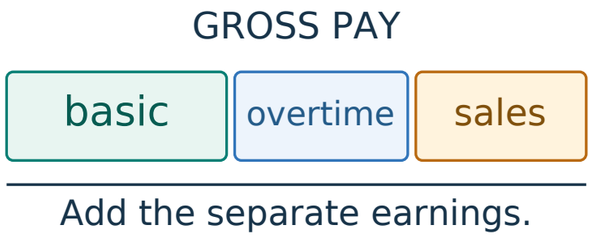
Add the bands + fixed charge.

Subtotal = fixed charge + Σ(band units × band rate).

Remember: Only the units inside a band receive that band’s rate.

04 / 4.3

Build gross pay from its parts



Gross pay = basic pay + overtime pay + commission.

Remember: Apply each hours or sales rate only within its own band.

05 / 4.3

Apply supplied tax bands and credits



Subtract all deductions from gross pay.

Gross tax = Σ(band income × rate)

income tax = max(0, gross tax − credits)

net = gross − all deductions.

Remember: Calculate USC and PRSI separately using the question’s rules.

06 / 4.3

Work backwards from net pay or tax paid

gross − deductions = net



Solve, then check the tax band.

Within one band: write net as a function of gross

Fully used credit = gross tax − income tax paid.

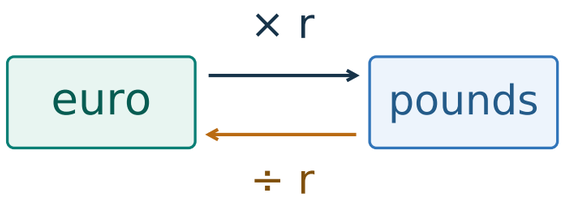
Remember: At zero income tax, an unused larger credit cannot be determined uniquely.

Applied Arithmetic

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

07 / 4.4

Convert currency in the right direction



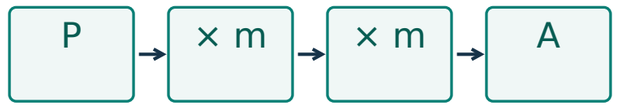
If €1 = £r: pounds = euro × r

euro = pounds/r, with r > 0.

Remember: Apply fees at the stated stage and in the stated currency.

08 / 4.5

Repeat a growth or depreciation multiplier



Repeat the multiplier n times.

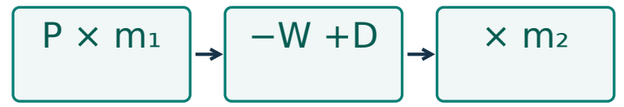
$A = P(1 + r/100)^n$

depreciation $A = P(1 - r/100)^n$.

Remember: Match rate and period; apply each change to the current amount.

09 / 4.5

Put rate changes and transactions in order



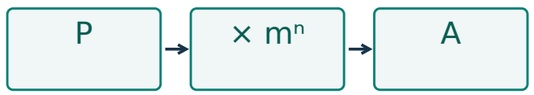
balance 1 → transaction → balance 2

$A = (P\,m_1 - W + D)m_2$.

Remember: W and D occur after period 1 here; follow the actual timing.

10 / 4.5

Recover the principal or repeated rate



Reverse with division or an nth root.

P = A/(m₁m₂…)

equal periods: m = (A/P)^(1/n)

Growth r = 100(m − 1)%

depreciation r = 100(1 − m)%.

Remember: Use positive balances; distinguish a multiplier from a percentage.

11 / 4.5

Tax the interest, not the principal



Keep interest × (1−t); principal stays.

Net interest = P × r × (1 − t), with decimal rates r and t.

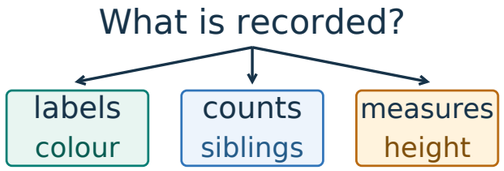
Remember: Interest tax applies to interest; retain the principal separately.

Statistics 1 - Collecting data

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 5.1

Choose the variable and data type



Population: target group

Variable: recorded feature

Categorical = labels

discrete = counts

continuous = measurements.

Remember: Rounding a measurement does not turn it into a count variable.

02 / 5.1

Repair a biased questionnaire

“How much do you love maths?”

↓

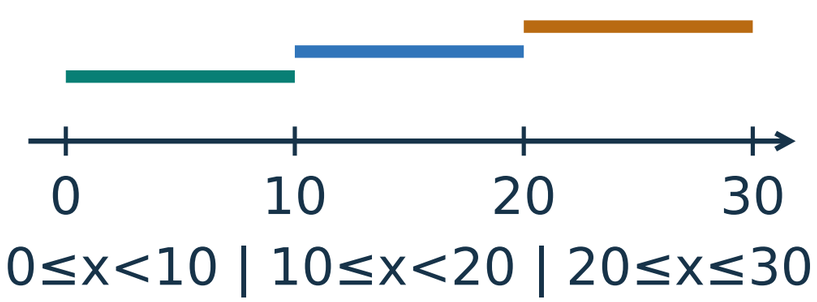
“How do you feel about maths?”

Use neutral wording, one idea at a time, and a precise reference period.

Remember: Avoid leading language and questions that ask two things at once.

03 / 5.1

Give every response exactly one home



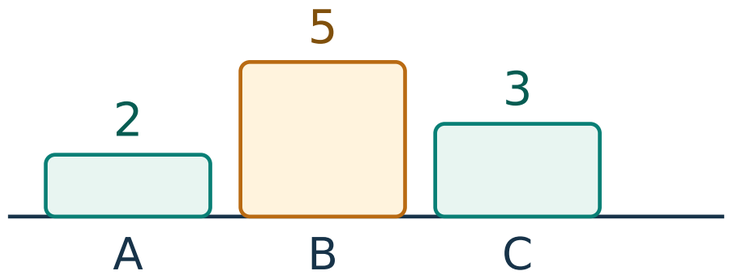
Response categories must be exhaustive and mutually exclusive.

Remember: Use inequalities or clear boundaries so endpoints belong once.

04 / 5.1

Turn counts into relative frequencies

N=10; B: 5/10 = 50%



Relative frequency = f/N

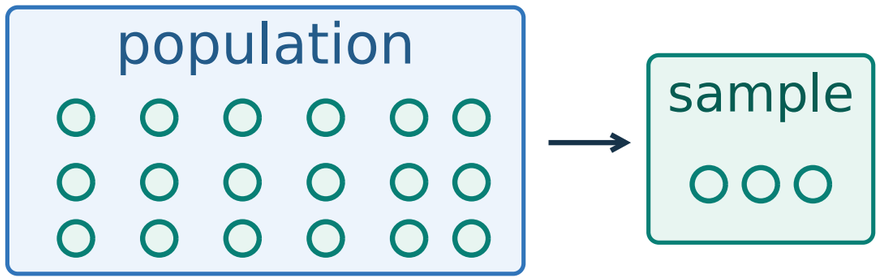
percentage = 100f/N.

Remember: N counts observations, not categories.

Statistics 1 - Collecting data

05 / 5.2

Check the source and sampling method



Primary: collected for this investigation

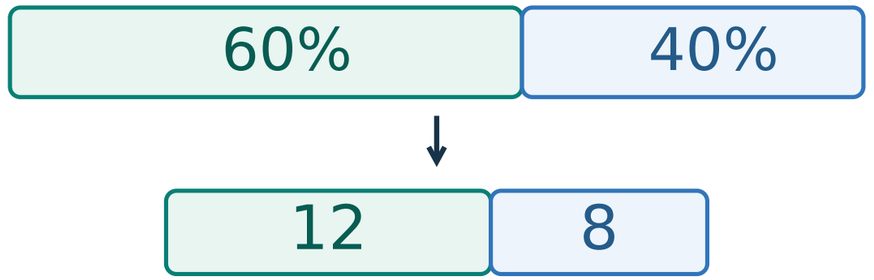
Secondary: already collected

Judge selection and coverage, not just size.

Remember: A large biased sample still describes the wrong group.

06 / 5.2

Take a proportional sample



Group sample = total sample × group population / total population.

Remember: After allocating quotas, select fairly within each group.

Perimeter - Area - Volume

01 / 6.1

Split a composite shape into familiar pieces



Area: add pieces. Perimeter: trace outside.

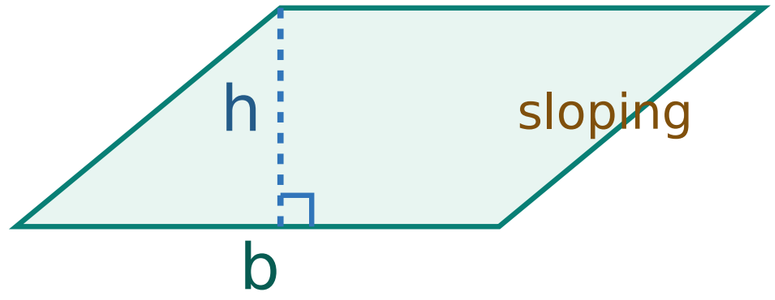
Rectangle: A = lw, P = 2(l + w)

Trapezium: A = (a + b)h/2. Kite/rhombus: A = $d_1d_2/2$. Add/subtract non-overlapping areas.

Remember: Trace only the outer boundary; h and the two diagonals must be perpendicular as required.

02 / 6.2

Use the perpendicular height



Parallelogram A = bh

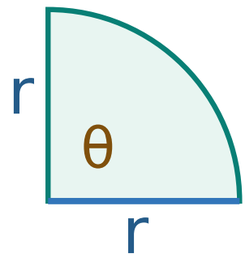
triangle A = bh/2.

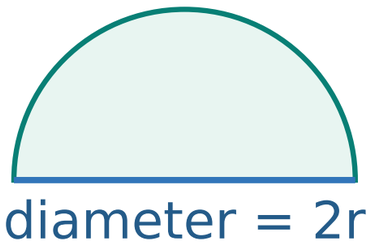
Remember: h meets the base at 90°; a sloping side is not the height.

Perimeter - Area - Volume

03 / 6.3

Choose a circular area or boundary formula





C = 2πr

A = πr²

Arc = (θ/360)2πr

sector A = (θ/360)πr²

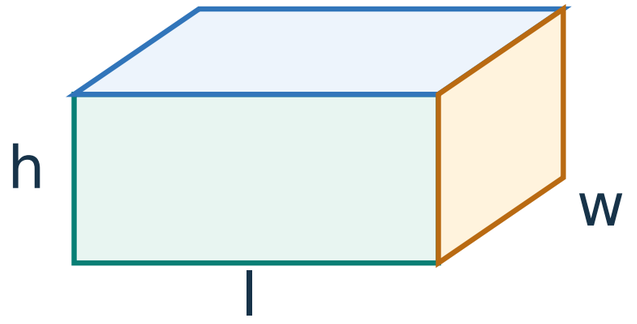
sector P = 2r + arc

Semicircle A = πr²/2, P = πr + 2r.

Remember: Sector 0<θ<360°. Count straight radii/diameters only when exposed.

04 / 6.4

Find cuboid volume or covering area



V = lwh

closed SA = 2(lw + lh + wh). 1 cm³ = 1 mL

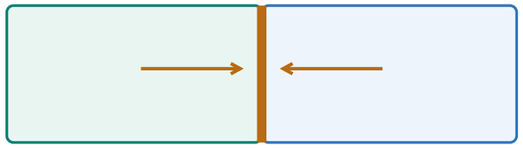
1000 cm³ = 1 L.

Remember: Volume fills the box; surface area covers its faces.

Perimeter - Area - Volume

05 / 6.4

Remove hidden faces when joining solids



two touching faces become internal

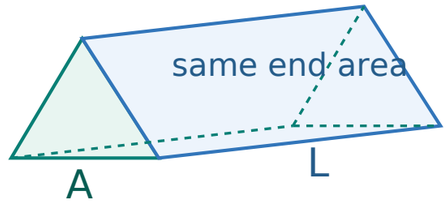
Exposed SA = $SA_1 + SA_2 - 2A_{shared}$

volumes add when solids do not overlap.

Remember: One full shared face is hidden on BOTH solids.

06 / 6.5

Multiply cross-section area by length



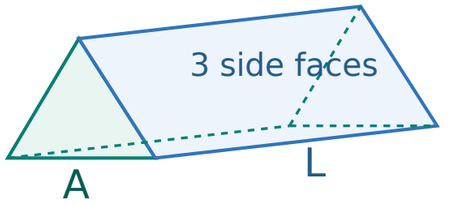
V = $A_{cross} \times L$

L = V/A_{cross}, where $A_{cross} > 0$.

Remember: Find the end area first; a triangular end needs the factor ½.

07 / 6.5

Cover a closed right prism

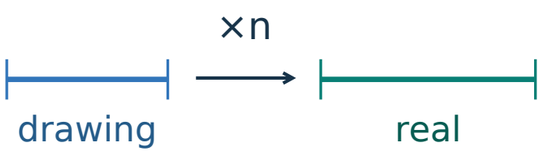


SA = $2A_{cross} + P_{cross} \times L$.

Remember: For a closed right prism: two ends plus all side rectangles.

08 / 6.6

Convert a map length both ways



Reverse the arrow by dividing by n.

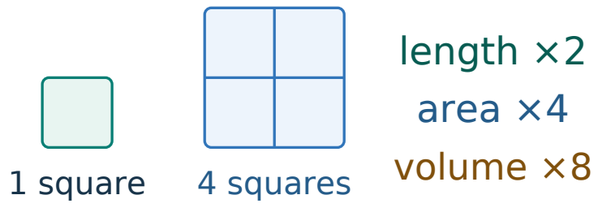
At scale 1:n: real = n × drawing

drawing = real/n.

Remember: Convert to the same units before applying the ratio.

09 / 6.6

Scale lengths, areas, volumes and costs



Length factor k → area factor k^2, volume factor k^3

Cost = quantity × unit price.

Remember: Use square or cubic factors for square or cubic units.

Statistics 2 - Averages and Variability

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 7.1

Find the mode and ordered median

2 | 4 | 4 | 7 | 9

middle = 4; most frequent = 4

Mode = most frequent

Odd N: position (N + 1)/2. Even N: mean of positions N/2 and N/2 + 1.

Remember: Sort first. There may be several modes or no unique mode.

02 / 7.2

Recover a total or missing observation

4 observations × mean 6 = total 24

3 | 5 | 7 | 9 ?

Mean = total/N

total = N × mean

missing = required total − known total.

Remember: N includes the missing observations.

03 / 7.2

Combine group means with their sizes

3 people mean 8 | 2 people mean 13

(3×8 + 2×13)/5 = 10

Combined mean = Σ(n × group mean)/Σn.

Remember: Weight each mean by its group size.

04 / 7.3

Choose an average that answers the question



An extreme pulls the mean towards it.

Mean: all values

Median: middle, resistant to extremes

Mode: most common value or category.

Remember: Justify the choice in context; no average is always best.

05 / 7.4

Weight values by their frequencies

x	f	fx
2	3	6
5	2	10

mean = (6+10)/(3+2) = 3.2

Mean = Σfx/Σf.

Remember: A value repeated f times contributes f×x to the total.

06 / 7.4

Estimate a mean from intervals

replace this interval by its midpoint



midpoint × frequency → estimated total

Midpoint = (lower + upper)/2

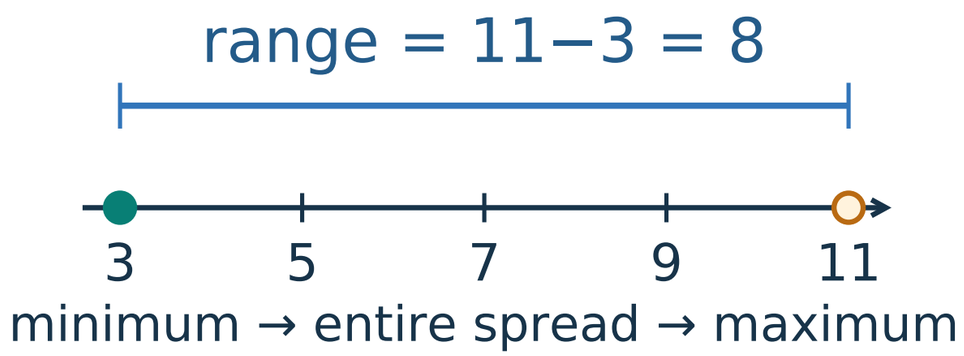
estimated mean = Σ(f × midpoint)/Σf.

Remember: The midpoint represents unknown individual values, so the mean is an estimate.

Statistics 2 - Averages and Variability

07 / 7.5

Use the range to find an extreme



Range = max − min

max = min + range

min = max − range.

Remember: Check the resulting minimum and maximum against every other value.

08 / 7.5

Check whether all statistics can hold



Every condition must fit the SAME data set.

Test total, ordering, median, range and integer conditions together

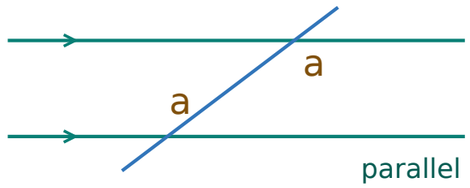
One contradiction rules the claim out.

Remember: A plausible average alone does not prove a data set exists.

Geometry 1 - Triangles, Quadrilaterals and Theorems

01 / 8.1

Find angles on lines and transversals



Straight = 180°

turn = 360°

vertical opposites equal

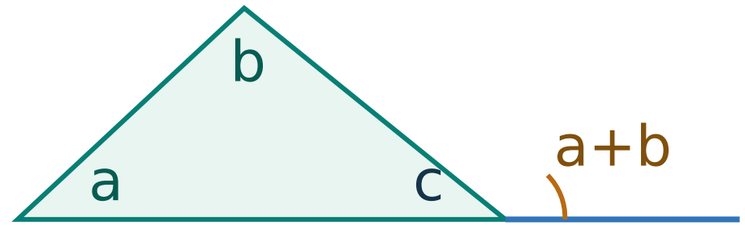
Parallel lines: corresponding/alternate equal

same-side interiors sum 180°.

Remember: State the parallel-line condition before using F, Z or interior-angle rules.

02 / 8.2

Link triangle interiors and an exterior angle



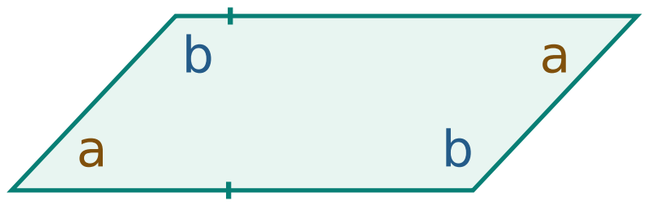
a + b + c = 180°

exterior angle = a + b for the two remote interiors.

Remember: The adjacent interior is not part of the remote-angle sum.

03 / 8.3

Use only the quadrilateral properties given



a+b=180°; 2a+2b=360°

Simple quadrilateral: interior sum = 360°

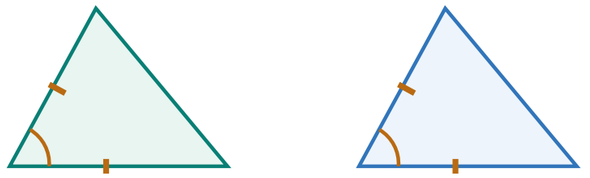
Parallelogram: opposite sides/angles equal

adjacent angles sum 180°.

Remember: A rhombus need not have right angles; do not add unstated properties.

04 / 8.4

Match triangles with a valid congruence test



SAS: angle between the marked sides

SSS

SAS (included angle)

ASA

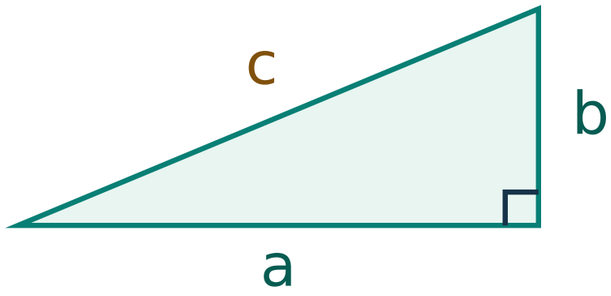
RHS (right triangles).

Remember: AAA gives similarity; SSA is not a general congruence test.

Geometry 1 - Triangles, Quadrilaterals and Theorems

05 / 8.5

Find a side in a right triangle



$c^2 = a^2 + b^2$

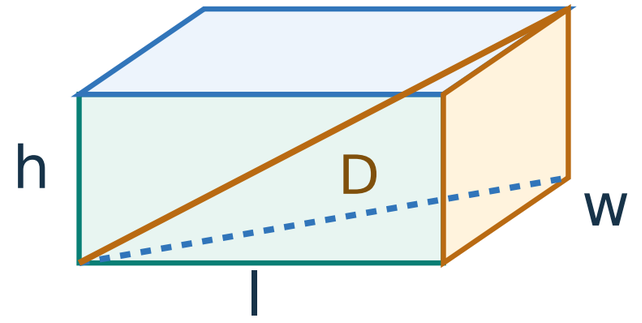
$c = \sqrt{(a^2 + b^2)}$

$a = \sqrt{(c^2 - b^2)}$.

Remember: c is opposite the 90° angle. Take the square root for a length.

06 / 8.5

Find a cuboid’s space diagonal



$d = \sqrt{(l^2 + w^2)}$

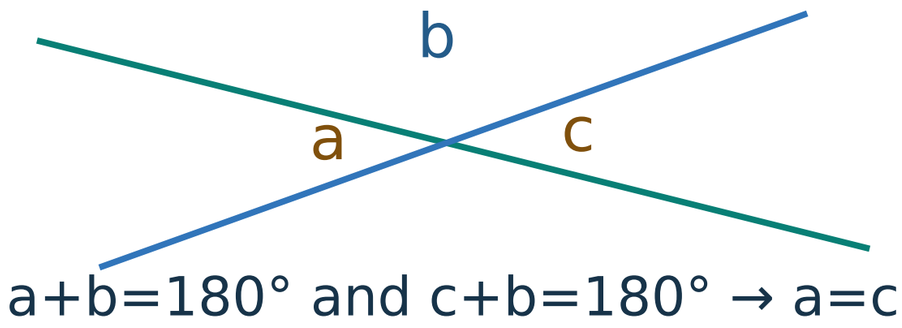
$D = \sqrt{(d^2 + h^2)} = \sqrt{(l^2 + w^2 + h^2)}$.

Remember: Use Pythagoras twice; an edge route is longer than the straight diagonal.

Geometry 1 - Triangles, Quadrilaterals and Theorems

07 / 8.6

Build a proof from accepted reasons



Definition = meaning

axiom = starting fact

theorem = proved fact

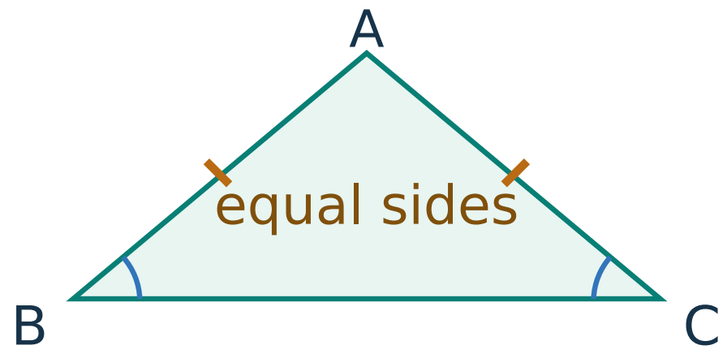
converse = reversed implication

corollary = direct consequence.

Remember: Equal-looking angles in a sketch do not prove equality.

08 / 8.6

Connect equal sides and opposite angles



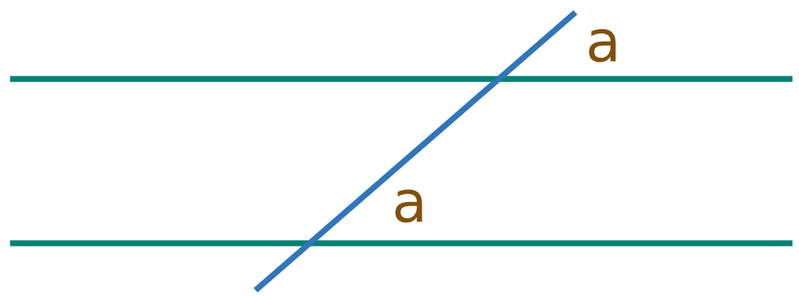
AB = AC ⇔ ∠ABC = ∠ACB in triangle ABC.

Remember: Forward theorem and converse require their own justification.

Geometry 1 - Triangles, Quadrilaterals and Theorems

09 / 8.6

Prove parallelism from matching angle positions



matching corresponding angles → parallel

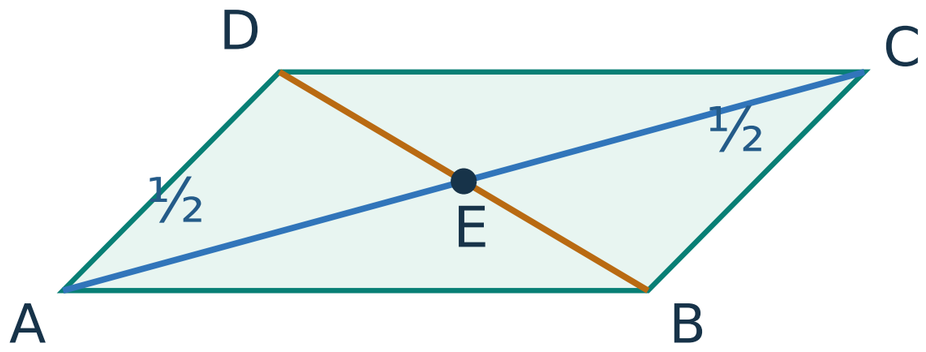
Equal alternate angles ⇒ parallel lines

Parallel lines ⇔ equal corresponding angles.

Remember: The equal angles must occupy the required transversal positions.

10 / 8.6

Use the bisecting diagonals of a parallelogram

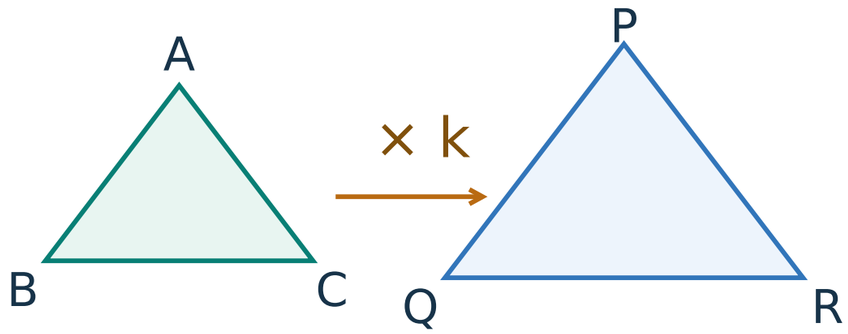


If diagonals AC and BD meet at E: AE = CE and BE = DE.

Remember: The two complete diagonals need not have equal lengths.

11 / 8.6

Match corresponding sides in similar triangles



AB/PQ = AC/PR = BC/QR

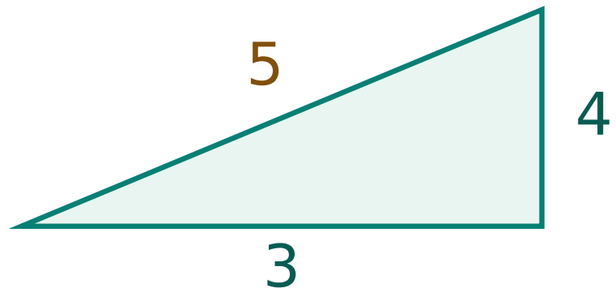
A ↔ P, B ↔ Q, C ↔ R.

Remember: Match corresponding vertices or opposite angles before forming ratios.

12 / 8.6

Test whether a triangle is right-angled

$3^2+4^2 = 5^2$



If c is longest and $c^2 = a^2 + b^2$, the angle opposite c is 90°.

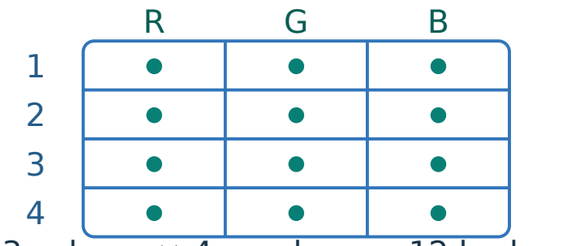
Remember: Put the longest side alone; this is the converse of Pythagoras.

Probability

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 9.1

Count all possible combinations



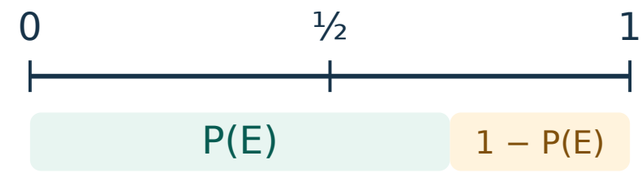
3 colours × 4 numbers = 12 badges

Total = $n_1 \times n_2 \times \ldots$

Remember: Multiply only when each earlier choice has the stated number of later choices.

02 / 9.2

Find the probability of “not”



Impossible ←→ Certain

$0 \leq P(E) \leq 1$

$P(\text{not } E) = 1 - P(E)$.

Remember: “Not both” includes exactly one and neither.

03 / 9.3

Count favourable outcomes



4 favourable / 12 equally likely

P(E) = favourable outcomes ÷ total outcomes

Remember: Use this count only when the individual outcomes are equally likely.

04 / 9.3

Update a mixture after adding items

18 red + 30 blue

48 existing	g added

New total: 48 + g

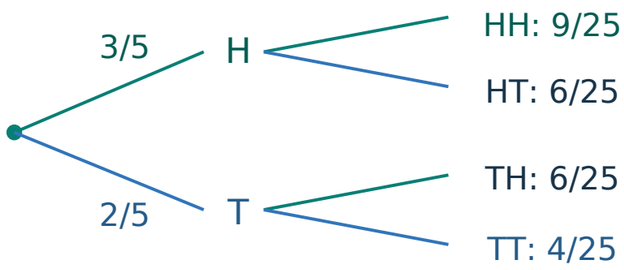
A:B = a:b ⇒ A = ak, B = bk

new total = old total + g.

Remember: An added share is measured against the final total.

05 / 9.4

Multiply along independent paths



$P(A \text{ and } B) = P(A)P(B)$

add mutually exclusive path products.

Remember: Use unchanged branch probabilities only for independent events.

06 / 9.3

Count repeated labels correctly



Three “2” faces out of six

P(2) = 3/6 = 1/2

Probability follows physical faces or sector angles.

Remember: Distinct labels need not have equal probabilities.

Probability

07 / 9.5

Estimate a future count

42 / 70
past trials

→

0.6 × 150
future trials

Estimate → expected count 90

Estimated P(E) = successes/trials

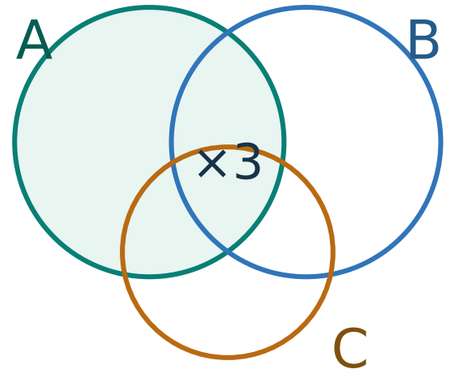
expected count = n × P(E).

Remember: An expectation assumes comparable conditions; it is not a guarantee.

08 / 9.6

Count overlapping sets once

Add
singles



− pairs
+ triple

n(A ∪ B) = n(A) + n(B) − n(A ∩ B)

n(A ∪ B ∪ C)

= n(A) + n(B) + n(C)

− n(A ∩ B) − n(A ∩ C) − n(B ∩ C)

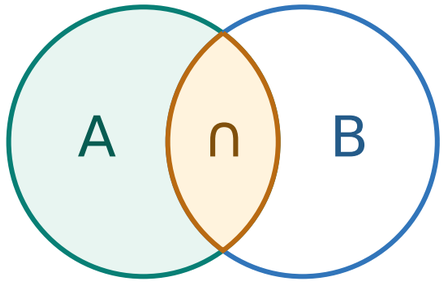
+ n(A ∩ B ∩ C)

Remember: Pair intersections include the triple overlap.

Probability

09 / 9.6

Turn words into set regions



Exactly one: leave out the overlap

AND: overlap OR: either circle

Or = union

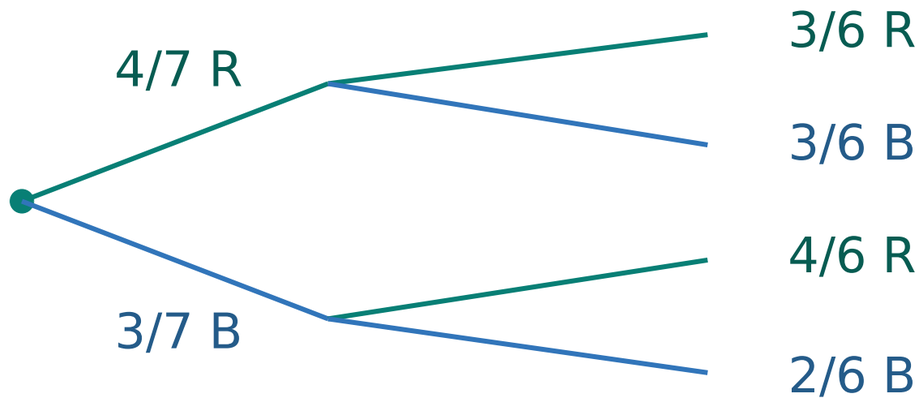
and = intersection

exactly one = single-set-only regions.

Remember: “At least two” includes the triple region in a three-set diagram.

10 / 9.7

Update a tree without replacement



P(A then B) = P(A) × P(B given A).

Remember: After removing one counter, update both counts and the total.

Simultaneous equations

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 10.1

Clear fractions from equations

$$\frac{x}{2} + \frac{y}{3} = 5$$

× 6 ↓

$$3x + 2y = 30$$

Multiply every term by the LCM of the denominators.

Remember: Denominators must be non-zero; multiply the right-hand side too.

02 / 10.1

Eliminate or substitute a variable

$$\begin{array}{r} 2x + y = 18 \\ x + y = 11 \\ \hline x = 7 \end{array}$$

subtract ↓

$y = 11 - 7 = 4$

Use equivalent operations

the answer must satisfy both equations.

Remember: Choose matching coefficients for elimination, or an isolated variable for substitution.

03 / 10.2

Build equations for two rates

6 hours × r (first rate) + 2 hours × s (second rate)

$$6r + 2s = \text{total pay}$$

Total = quantity₁ × rate₁ + quantity₂ × rate₂.

Remember: Define unknown rates and keep their units consistent.

04 / 10.2

Separate fixed cost and daily rate

F (once) + d + d + d (3 daily charges)

$$C = F + 3d$$

C = F + nd

two different n-values identify F and d.

Remember: The fixed charge occurs once, not once per day.

05 / 10.2

Translate changed ages or fractions

x − k → x now

k years later

Changed fraction: (n+a)/(d+b)

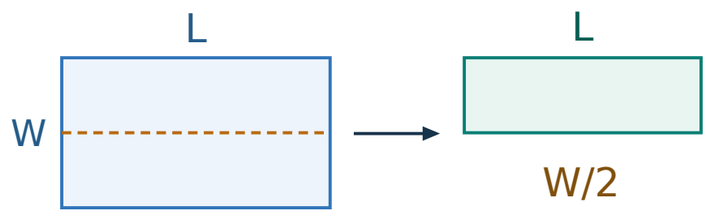
k years ago: x − k

changed fraction: (n + a)/(d + b).

Remember: Require d+b≠0. Twice a past age is 2(x−k).

06 / 10.2

Find dimensions after a fold



2L + 2W → 2L + W

P = 2(L + W)

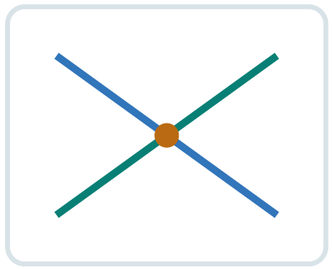
halving W gives P_folded = 2L + W.

Remember: A fold changes one dimension; it need not halve the perimeter.

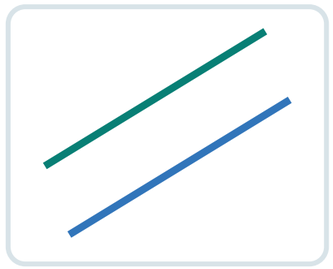
Simultaneous equations

07 / 10.2

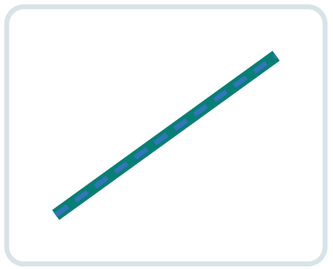
Recognise how many solutions exist



One pair



No pair



Infinite

Crossing: 1 solution

distinct parallel: 0

coincident: infinitely many.

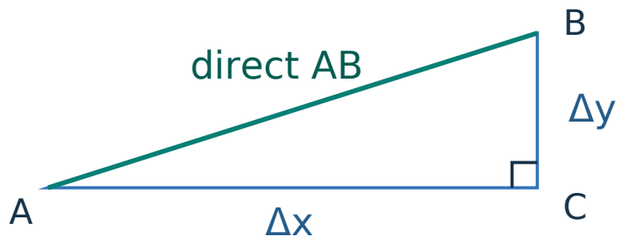
Remember: A simultaneous solution is a point on both lines.

Coordinate Geometry - The Line

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 11.1

Find a direct or multi-leg distance



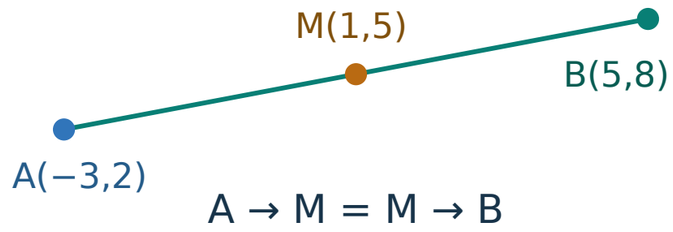
$AB = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

via C: AC + CB.

Remember: Use the distance formula for each straight leg of a route.

02 / 11.1

Find a midpoint or missing endpoint



A → M = M → B

M = ((x₁ + x₂)/2,

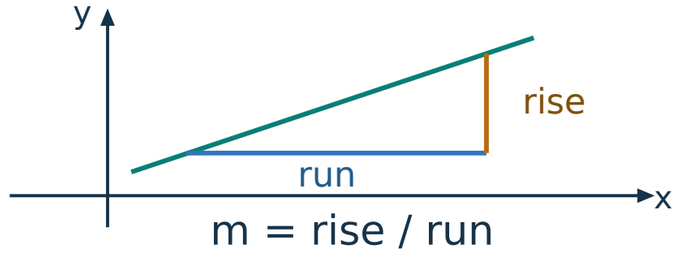
(y₁ + y₂)/2)

B = 2M − A.

Remember: Apply the missing-endpoint rule separately to each coordinate.

03 / 11.2

Find a slope and its exceptions



m = rise / run

$m = (y_2 - y_1)/(x_2 - x_1)$

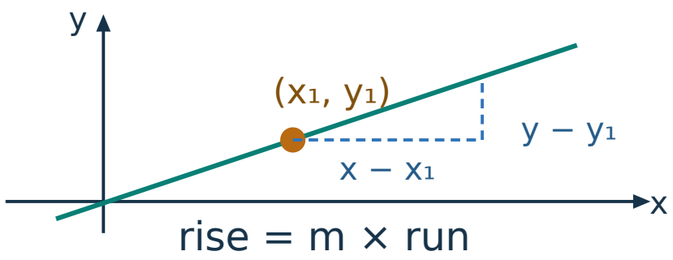
horizontal m = 0

vertical undefined.

Remember: Require $x_2 \neq x_1$ for the fraction; keep subtraction order consistent.

04 / 11.3

Build a line from a point and slope



rise = m × run

$y - y_1 = m(x - x_1)$.

Remember: Keep coordinates grouped: subtracting a negative becomes addition.

05 / 11.4

Read slope and intercept

$$3x + 2y - 8 = 0$$

$$\downarrow$$

$$y = -(3/2)x + 4$$

slope −3/2 intercept 4

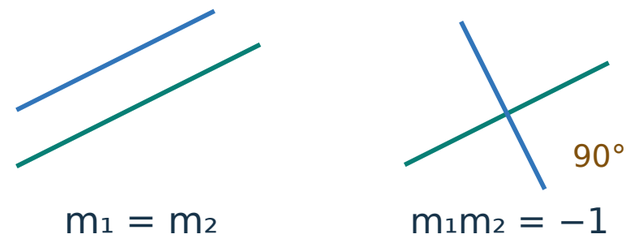
ax + by + c = 0 ⇒ y = −(a/b)x − c/b

m = −a/b.

Remember: Require b≠0. Isolate y before reading off coefficients.

06 / 11.5

Choose parallel or perpendicular slope



$m_1 = m_2$ $m_1 m_2 = -1$

Parallel: $m_1 = m_2$

perpendicular finite slopes: $m_1 m_2 = -1$. Horizontal y = k has m = 0

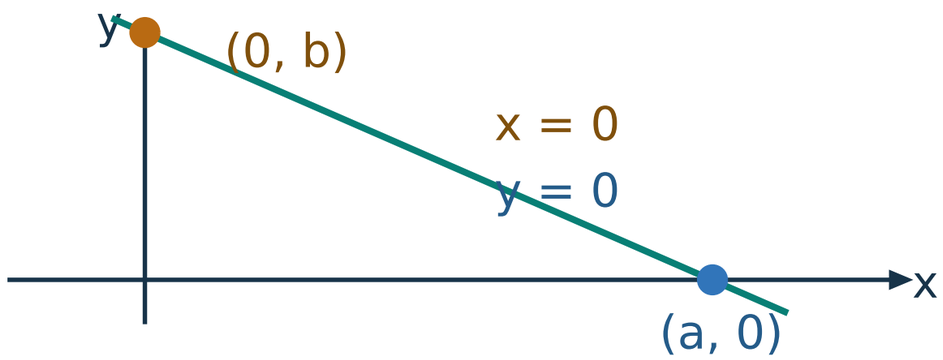
vertical x = k is undefined.

Remember: A horizontal-vertical pair is perpendicular; do not multiply undefined slopes.

Coordinate Geometry - The Line

07 / 11.6

Plot a line using intercepts



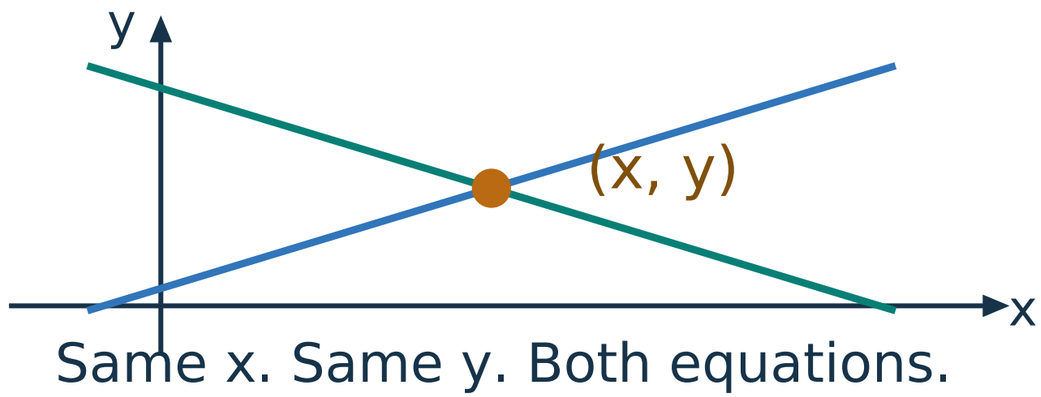
x-intercept: set y = 0

y-intercept: set x = 0.

Remember: If both give the origin, choose a different second point.

08 / 11.7

Find the intersection of two lines

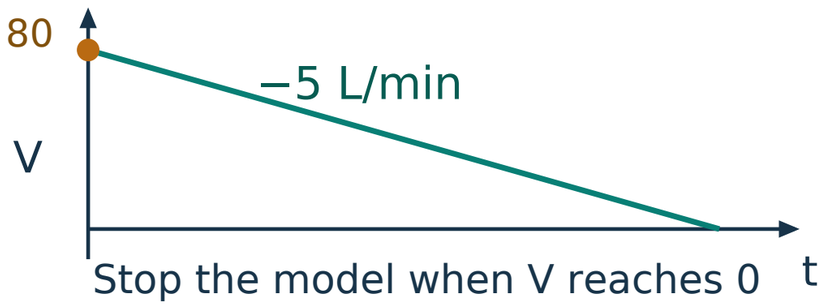


Solve the two line equations simultaneously.

Remember: Graph readings are approximate until confirmed algebraically.

09 / 11.8

Read a rate and sensible domain



y = mx + c

m = Δy/Δx

c = value at x = 0.

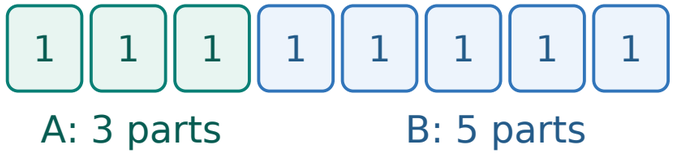
Remember: State units and restrict the model to meaningful inputs.

Ratio - Time - Speed

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 12.1

Share a total or rebuild it from one share



Total = 8 equal parts

A:B = a:b ⇒ A = aN/(a + b)

B = bN/(a + b)

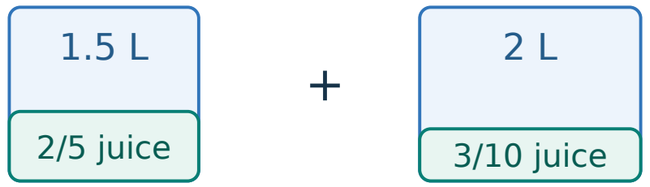
If A is known: one part = A/a

N = (a + b)A/a.

Remember: N is the total. For reconstruction require a≠0.

02 / 12.1

Mix batches with different ratios



Add ingredient amounts, then form a ratio

Ingredient amount = batch volume × ingredient parts/total parts.

Remember: Find ingredient amounts before combining; do not average ratios.

03 / 12.1

Distinguish direct and inverse proportion



Compare the constant relationship

Direct: y = kx

Inverse: xy = k

Equal workers × time = fixed work.

Remember: Worker-time models assume equal productivity and divisible work.

04 / 12.1

Finish a job after the team changes

90 worker-days total



66 ÷ 11 workers = 6 more days

4 elapsed + 6 more = 10 days

Remaining worker-days = total worker-days − completed worker-days.

Remember: Use the remaining work for the new team; then add elapsed time.

05 / 12.2

Measure a duration across midnight



18 + 77 = 95 minutes

1 h = 60 min

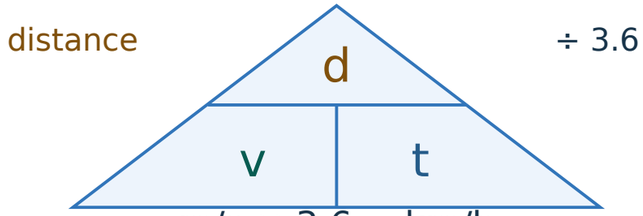
1 day = 24 h

Add 24 h to a next-day clock time before subtraction.

Remember: Put both times on one continuous timeline.

06 / 12.3

Choose the distance, speed or time formula



v = d/t

d = vt

t = d/v

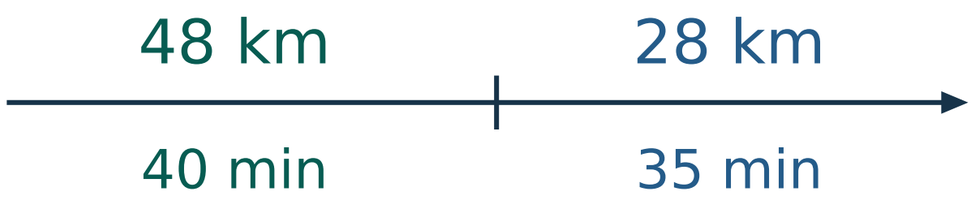
1 m/s = 3.6 km/h.

Remember: Match time and distance units; 45 min=0.75 h.

Ratio - Time - Speed

07 / 12.3

Find the average speed of a journey



76 km ÷ 1.25 h = 60.8 km/h

Average speed = total distance/total elapsed time.

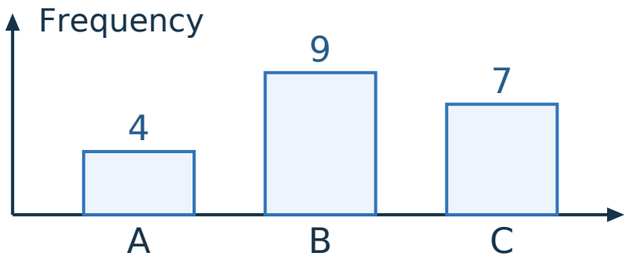
Remember: Include stops unless moving time is specifically requested.

Statistics 3 - Presenting Data

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 13.1

Read categories and frequencies



Each dot = one observation

labelled frequency-bar height = frequency.

Remember: A high bar means a frequent category, not a larger data value.

02 / 13.1

Recover a missing value from a mean

known sum: 216 | ?

10 values × mean 25 = 250

missing = 250 − 216 = 34

Mean = Σvalues/n

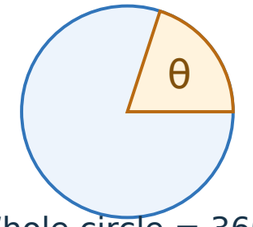
total = n × mean

missing = n × mean − known sum.

Remember: Use the final number of values, including the missing one.

03 / 13.2

Convert a pie angle to a share or total



Whole circle = 360°

$$\frac{\text{part f}}{\text{whole N}} = \theta / 360°$$

θ = 360°f/N

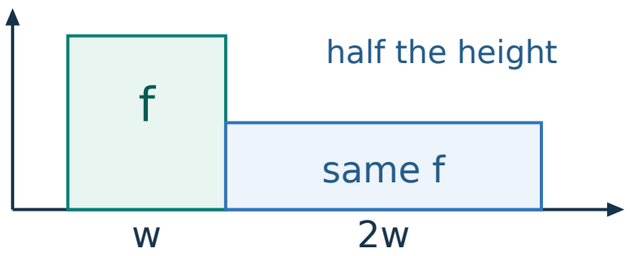
f = (θ/360°)N

N = 360°f/θ.

Remember: Use non-zero N; recovering N also requires θ≠0.

04 / 13.3

Read frequency from a histogram



Density = f/w

frequency = bar area = density × class width.

Remember: With unequal class widths, compare areas rather than heights.

05 / 13.4

Read a stem-and-leaf display

Stem	Leaves
1	2 8 8
2	3 5 5

Key 1|2=12

Middle values: 18 and 23

Median: position (n + 1)/2 if n odd

average positions n/2 and n/2 + 1 if even

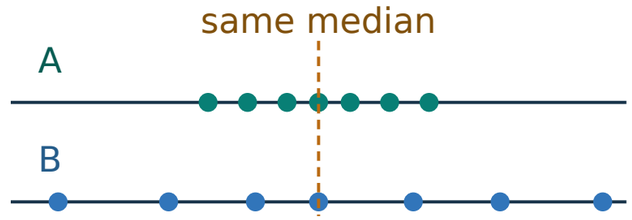
Mode = most frequent

range = max − min.

Remember: Read the place-value key and put leaves in order.

06 / 13.4

Compare centre and spread



A has the smaller range

Median compares centres

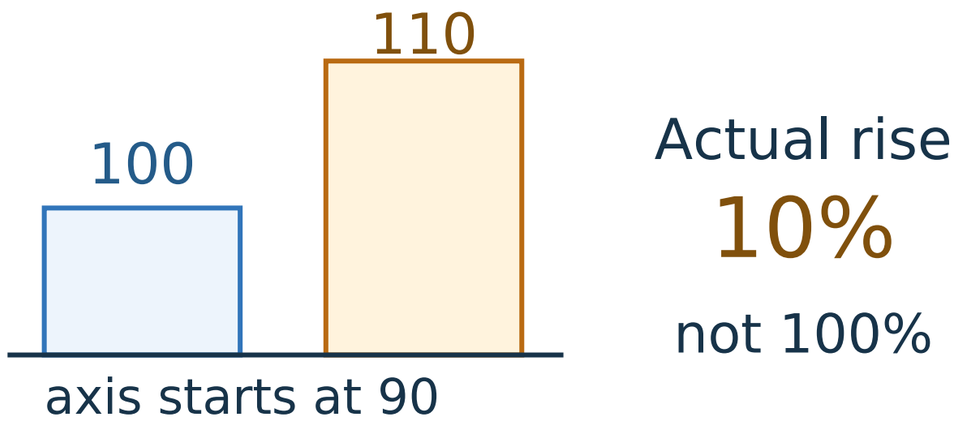
range = max − min compares overall spread.

Remember: A smaller range indicates greater consistency by that measure only.

Statistics 3 - Presenting Data

07 / 13.5

Check whether a chart misleads



Percentage change = (new − old)/old × 100%

A k-by-k pictogram has area factor k^2.

Remember: Require old≠0. Check baselines, scales and pictogram areas.

Indices - Standard form - Surds

01 / 14.1

Choose the correct index law

$$x^3 \times x^2$$

x × x × x × x × x

x × x × x

5 factors of x → x^5

- $a^m a^n = a^{m+n}$
- $a^m / a^n = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $(ab)^n = a^n b^n$
- $(a/b)^n = a^n / b^n$.

Remember: Exclude zero in divisors/negative powers. Use positive bases for rational-index laws.

02 / 14.2

Interpret zero, negative and fractional powers

a^0

1

a^{-n}

$1/a^n$

a^(m/n)

root → power

Denominator: root. Numerator: power.

- $a^0 = 1$
- $a^{-n} = 1/a^n$
- a^(m/n) = $(\sqrt[n]{a})^m$.

Remember: For zero/negative powers a≠0. For the stated fractional rule, a>0 and n is a positive integer.

Indices - Standard form - Surds

03 / 14.3

Solve an equation with equal bases

$$3^{(2x-1)} = 27^{(x-2)}$$

$$3^{(2x-1)} = 3^{[3(x-2)]}$$

$$2x - 1 = 3(x - 2)$$

Same base → match exponents

For $b > 0$, $b \neq 1$: $b^u = b^v \Rightarrow u = v$.

Remember: Changing 27 to 3^3 multiplies the whole outer exponent by 3.

04 / 14.4

Simplify and combine matching surds

$\sqrt{72}$ → $\sqrt{36} \times \sqrt{2}$

$$6\sqrt{2}$$

Take out the largest square factor

$\sqrt{(ab)} = \sqrt{a}\sqrt{b}$

$p\sqrt{q} + r\sqrt{q} = (p + r)\sqrt{q}$.

Remember: Use $a,b \geq 0$. Combine only matching simplified square-root parts.

05 / 14.4

Multiply surds or clear a root denominator

$$\frac{3}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

Multiply numerator AND denominator

$\sqrt{a}\sqrt{b} = \sqrt{(ab)}$

$\sqrt{a}/\sqrt{b} = \sqrt{(a/b)}$

$p/\sqrt{q} = p\sqrt{q}/q$

$(\sqrt{a} \pm \sqrt{b})^2 = a + b \pm 2\sqrt{(ab)}$.

Remember: $a,b \geq 0$ for products; $b>0$ for quotients; $q>0$ for rationalising.

06 / 14.5

Multiply or divide standard-form numbers

$(6.4 \div 2.5) \times 10^{(9-6)}$

↓

$$2.56 \times 10^3$$

$1 \leq |\text{coefficient}| < 10$

Product = $ab \times 10^{m+n}$

quotient = $(a/b) \times 10^{m-n}$, $b \neq 0$.

Remember: Normalise a non-zero answer so $1 \leq |\text{coefficient}| < 10$.

07 / 14.5

Add or subtract standard-form numbers

$$6.4 \times 10^9 - 7.5 \times 10^8$$

$$6.4 \times 10^9 - 0.75 \times 10^9$$

$$(6.4 - 0.75) \times 10^9$$

Match powers before adding coefficients

Write a common power of ten

then add or subtract coefficients.

Watch for: Do not add or subtract exponents for addition or subtraction.

08 / 14.6

Round to the requested accuracy

0.00 6 2 84 →

leading zeros keep 2 digits 8 rounds up

0.006284 → 0.0063 (2 s.f.)

Decimal places count from the decimal point

significant figures start at the first non-zero digit.

Remember: Use the next digit to decide rounding; retain place value.

Indices - Standard form - Surds

09 / 14.7

Enter brackets and test a claim

$(3 + 4)^2$

49

$\neq$

$3^2 + 4^2$

25

One counterexample disproves “always”

Bracket compound numerators, denominators and radicands

One valid counterexample disproves a universal claim.

Remember: One agreeing example cannot prove an identity for all values.

Quadratic equations

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 15.1

Solve by factorising

$$x^2 - x - 12 = 0$$

$$(x - 4)(x + 3) = 0$$

$x = 4$ $\qquad$ $x = -3$

> $uv = 0 \Rightarrow u = 0$ or $v = 0$.

Remember: First rearrange the equation so the product equals zero.

02 / 15.2

Use the quadratic formula

$$3x^2 - 5x - 2 = 0$$

$a = 3$ $\qquad$ $b = -5$ $\qquad$ $c = -2$

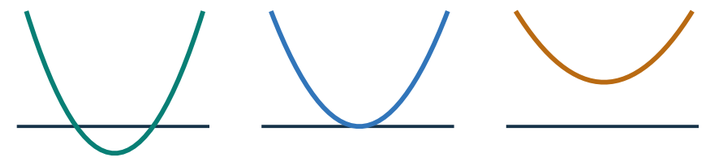
Keep the sign with each coefficient

> $x = [-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$, $a \neq 0$.

Remember: Write $ax^2+bx+c=0$ first; keep the signs of a, b and c.

03 / 15.2

Predict how many real roots exist



D > 0
2 roots

D = 0
1 repeated

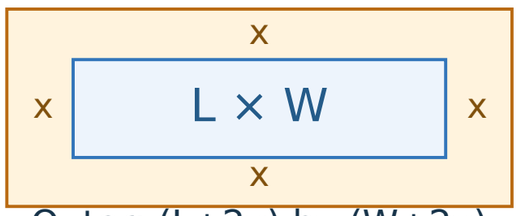
D < 0
0 real roots

> $D = b^2 - 4ac$: $D > 0$ → two
>
> $D = 0$ → repeated
>
> $D < 0$ → no real roots.

Remember: A negative D means no real roots, not two negative roots.

04 / 15.3

Find a uniform border width



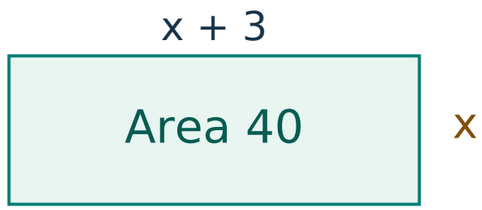
Outer: (L+2x) by (W+2x)

> Outer area = $(L + 2x)(W + 2x)$
>
> border area = outer area − LW.

Remember: A border on both sides adds 2x to each dimension.

05 / 15.3

Build and check a quadratic model



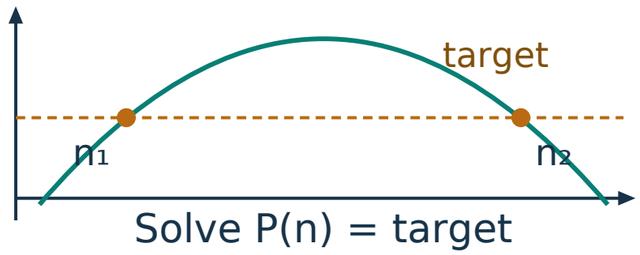
$x(x+3)=40 \rightarrow x>0$

> Translate the original relation → solve → apply the domain.

Remember: Reject a root only when it violates a stated or physical restriction.

06 / 15.3

Find both times a target is reached



Solve P(n) = target

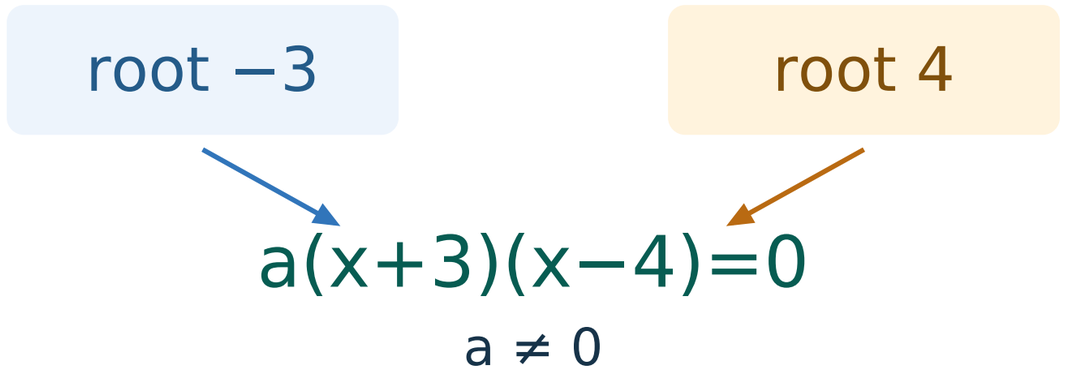
> Set model expression = target, then rearrange to zero and solve.

Remember: Both roots may be valid; test them against the model's domain.

Quadratic equations

07 / 15.4

Build an equation from its roots



Roots r, s: $a(x - r)(x - s) = 0$

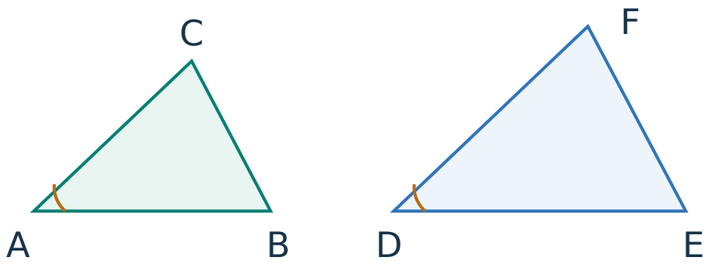
expand to $ax^2 - a(r + s)x + ars = 0$, $a \neq 0$.

Remember: A negative root creates a plus sign in its factor.

Geometry 2 - Similar Triangles, Circles and Theorems

01 / 16.1

Match sides of similar triangles

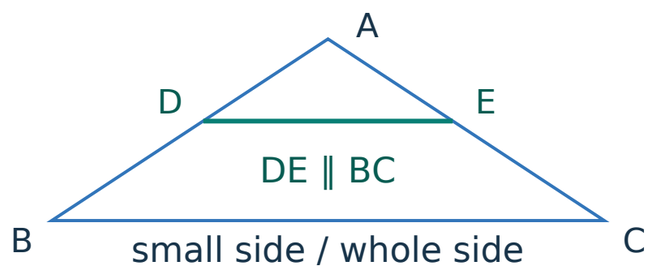


AA: two equal angle pairs suffice. ΔABC~ΔDEF ⇒ AB/DE = BC/EF = AC/DF.

Remember: Corresponding angles are equal; all matching lengths use one scale.

02 / 16.2

Use a parallel cut in a triangle



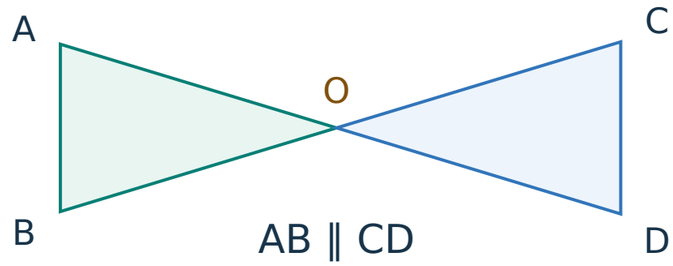
DE∥BC ⇒ AD/AB = AE/AC = DE/BC

AD/DB = AE/EC.

Remember: Match whole with whole or remaining part with remaining part; denominators ≠0.

03 / 16.2

Match crossing similar triangles

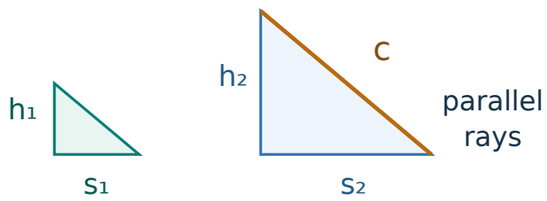


Vertically opposite angles + a parallel-line angle pair ⇒ AA similarity.

Remember: Match vertices before taking ratios, even when a triangle is reversed.

04 / 16.1

Use shadows and a right triangle



$h_1/s_1 = h_2/s_2$

$c^2 = a^2 + b^2$

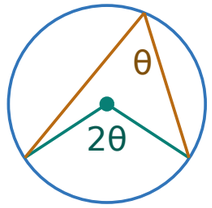
$c = \sqrt{(a^2 + b^2)}$

$a = \sqrt{(c^2 - b^2)}$.

Remember: Shadow ratios assume parallel sun rays on level ground; c is the hypotenuse.

05 / 16.3

Choose the correct circle-angle fact



Same arc: 2θ

Diameter: 90°

Cyclic: α+β=180°

Same arc: centre = 2 × edge

Same segment: edge angles equal

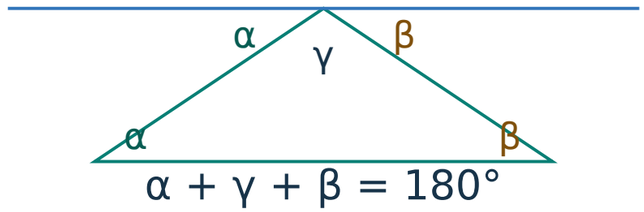
Diameter: edge = 90°

Cyclic opposite angles sum to 180°.

Remember: Identify the same arc; the relevant centre angle may be reflex.

06 / 16.4

Prove the triangle-angle sum



α + β + γ = 180° in a Euclidean triangle.

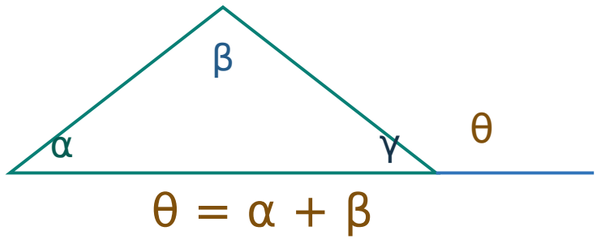
Remember: Use an auxiliary parallel line and known angle facts, not the theorem being proved.

Geometry 2 - Similar Triangles, Circles and Theorems

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

07 / 16.4

Prove the exterior-angle theorem



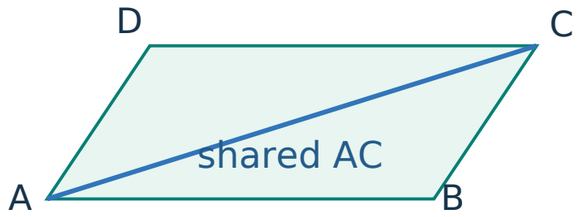
$\theta = \alpha + \beta$

Exterior angle θ = α + β, the two non-adjacent interior angles.

Remember: Do not include the adjacent interior angle in the remote-angle sum.

08 / 16.4

Prove opposite parallelogram parts equal



parallel angles + common side → ASA

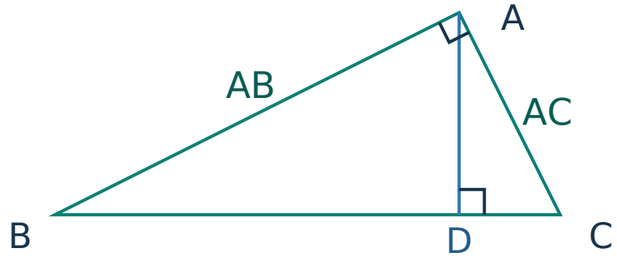
AB∥CD and BC∥AD

diagonal AC gives ΔABC≅ΔCDA by ASA.

Remember: A parallelogram diagonal need not bisect each corner angle.

09 / 16.4

Prove Pythagoras with an altitude



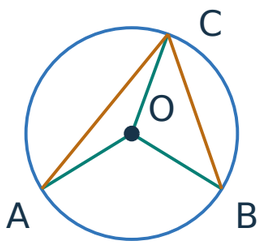
$AB^2 + AC^2 = BC^2$

BC is the hypotenuse.

Remember: Drop AD perpendicular to BC; D need not be its midpoint.

10 / 16.4

Prove the centre-angle theorem



$OA = OB = OC$

equal base angles

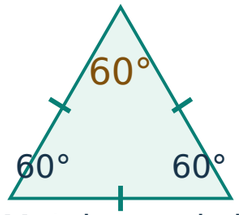
centre = 2 × edge

Central angle on the same arc = 2∠ACB.

Remember: Keep the chosen arc fixed; some cases use a reflex central angle.

11 / 16.1

Use equal sides and angle structure





Match equal sides to opposite angles

Isosceles: equal sides ⇒ equal opposite angles

Equilateral: all sides equal, angles 60°

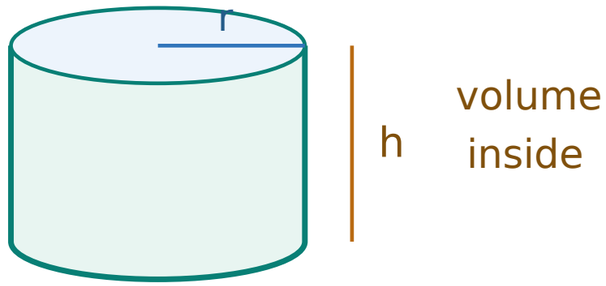
Parallelogram: opposite parts equal, adjacent angles sum to 180°.

Remember: Equal adjacent parallelogram sides alone do not make equilateral triangles.

Cylinder - Sphere

01 / 17.1

Find a cylinder’s volume, height or radius



$V = \pi r^2 h$

$h = V/(\pi r^2)$

$r = \sqrt{[V/(\pi h)]}$, with $r, h > 0$.

Watch for: Using diameter as radius, or taking a square root before isolating r^2.

02 / 17.1

Choose the cylinder surfaces



curved surface + present ends

Curved $A = 2\pi rh$

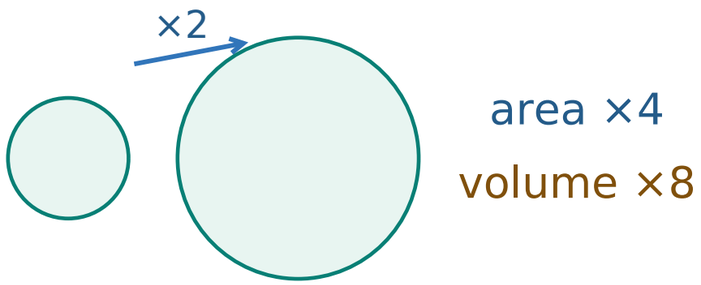
open top with base: $A = 2\pi rh + \pi r^2$

closed: $A = 2\pi rh + 2\pi r^2$.

Watch for: Adding a missing lid or confusing surface area with capacity.

03 / 17.1

Predict the effect of scaling



Cylinder $V \propto r^2 h$

sphere $V \propto r^3$

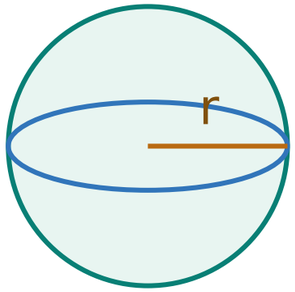
With h fixed, r × k gives V × k^2.

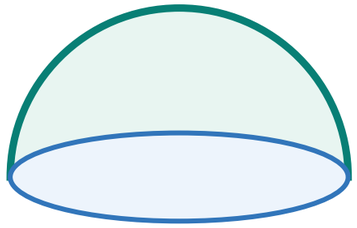
Remember: Cylinder height fixed: radius ×k means volume ×k^2. Sphere: volume ×k^3.

Cylinder - Sphere

04 / 17.2

Sphere or hemisphere: area or volume?





include base?

Sphere: $V = 4\pi r^3/3$

$A = 4\pi r^2$

Hemisphere: $V = 2\pi r^3/3$

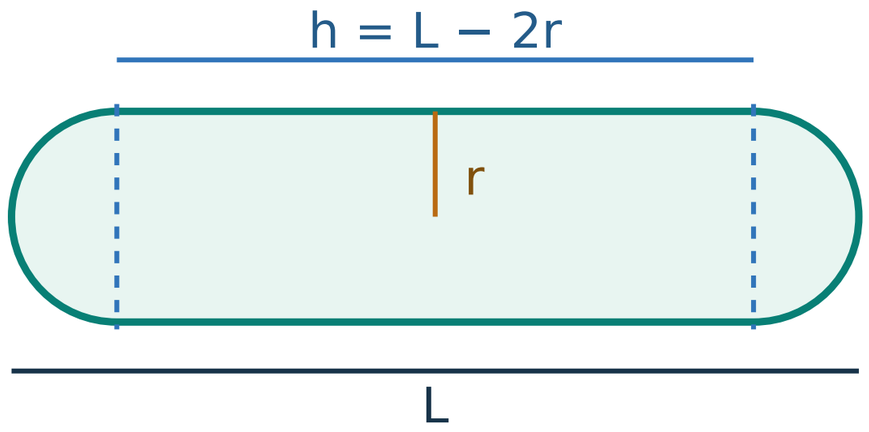
curved $A = 2\pi r^2$

closed $A = 3\pi r^2$.

Watch for: Adding a base at a hemisphere-cylinder join.

05 / 17.2

Split a capsule into familiar solids



For total end-to-end length L and radius r: $h = L - 2r$

$V = \pi r^2 h + 4\pi r^3/3$

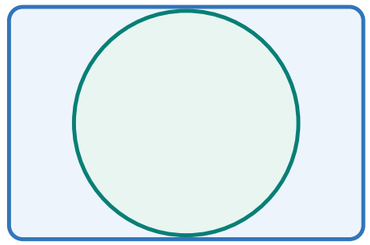
external $A = 2\pi rh + 4\pi r^2$, $L > 2r$.

Watch for: Using L as the cylinder height.

Cylinder - Sphere

06 / 17.2

Find empty space or complete litres



container

− solid

= space

Empty volume = container volume − enclosed solid volume

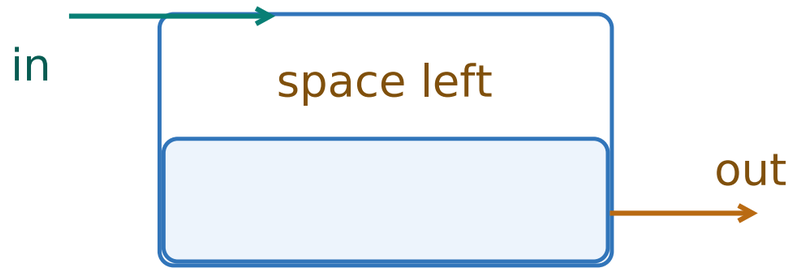
1 L = 1000 cm³

complete litres = floor(capacity in L).

Watch for: Rounding 3.9 litres up to four complete litres.

07 / 17.3

Find the time left to fill a tank



t = remaining volume / net inflow

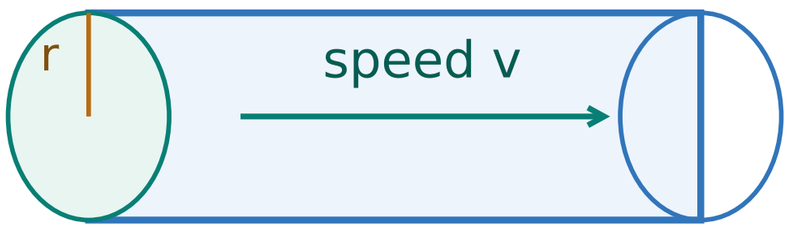
net inflow = inflow − outflow

use compatible units.

Watch for: Using full capacity when the vessel is partly full.

08 / 17.3

Turn pipe speed into volume flow



swept volume each second

Q = cross-sectional area × speed = πr²v

For cm and s, Q is cm³/s.

Watch for: Treating a speed in cm/s as though it were a flow in litres/s.

Cylinder - Sphere

09 / 17.3

Convert volume and time units



$$1\ \text{L} = 1000\ \text{cm}^3 = 1000\ \text{mL}$$

$1\ \text{m}^3 = 10^6\ \text{cm}^3 = 1000\ \text{L}$

$1\ \text{L} = 1000\ \text{cm}^3$

$1\ \text{mL} = 1\ \text{cm}^3$

$1\ \text{min} = 60\ \text{s}$

Area conversion squares a length factor

volume conversion cubes it.

Watch for: Using 1 m^3=100 cm^3 by applying only the linear conversion factor.

Patterns and Sequences

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 18.1

Separate position from term value



Input a position, not the previous term.

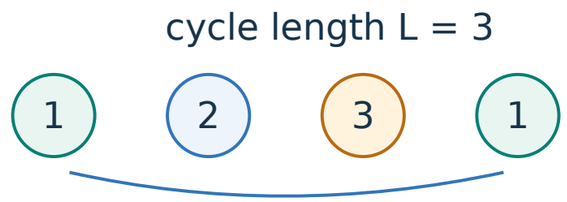
T_n is the value at position n

n is not generally the term value.

Watch for: Feeding a previous term into an nth-term rule.

02 / 18.2

Find a position in a repeating cycle



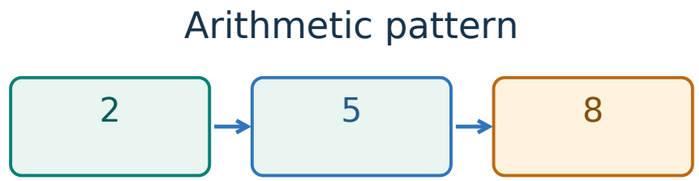
$n = qL + r,\ 0 \le r < L$

use cycle item r, except r = 0 means item L.

Watch for: Using zero remainder as the first item.

03 / 18.3

Recognise arithmetic or geometric growth



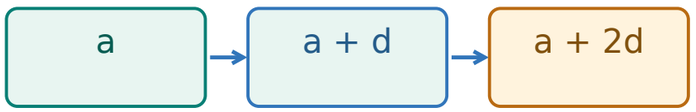
Arithmetic: $T_{n+1} - T_n = d$

Geometric: $T_{n+1}/T_n = r$ where the denominator is non-zero.

Watch for: Calling every increasing pattern linear.

04 / 18.4

Build or reverse an arithmetic rule



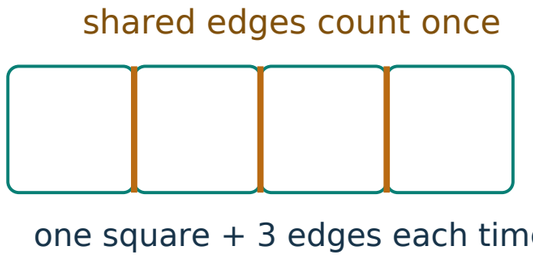
$T_n = a + (n - 1)d$

for $d \neq 0$, $n = 1 + (T - a)/d$.

Watch for: Accepting n = 14.2 as a position in an integer-indexed sequence.

05 / 18.5

Count a growing shape efficiently



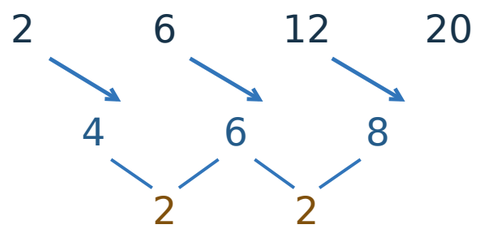
Total = fixed part + repeated part

count a shared edge once.

Watch for: Using the number of sides per shape without allowing for shared sides.

06 / 18.6

Find a quadratic nth-term rule



$T_n = an^2 + bn + c$

constant second difference = 2a for unit-spaced n.

Watch for: Using the second difference itself as a.

Patterns and Sequences

07 / 18.6

Find the position of a quadratic term



Check each candidate in the original rule.

Solve $an^2 + bn + c$ = target, then retain allowed integer n

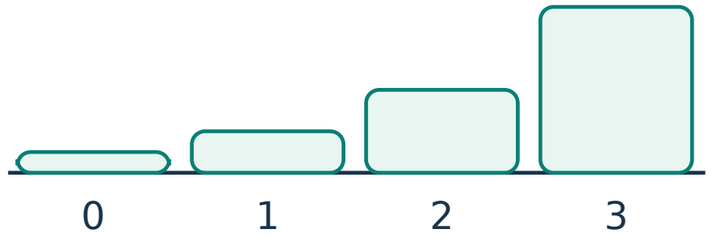
Three independent term conditions determine a general quadratic.

Watch for: Assuming two starting values uniquely determine a quadratic sequence.

08 / 18.7

Choose the exponent from the starting stage

a → 2a → 4a → 8a



If stage 1 is a: $T_n = ar^{n-1}$

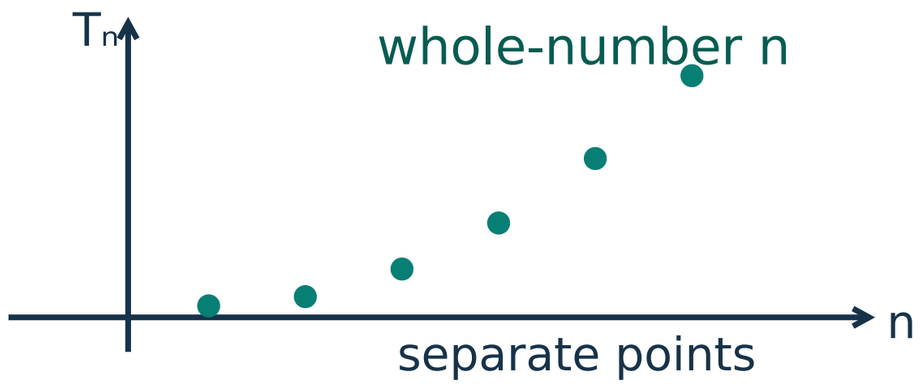
If stage 0 is a: $T_n = ar^n$

For positive a, r, $T_{n+k}/T_n = r^k$.

Watch for: Using ar^n when the supplied amount belongs to stage 1.

09 / 18.7

Plot a sequence as separate points



Plot (n, T_n) at allowed integer n

linear, quadratic and exponential growth have different difference patterns.

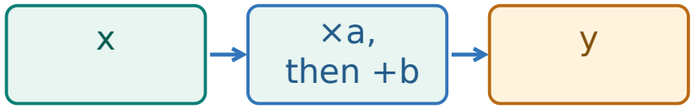
Watch for: Joining sequence points to suggest non-integer stages exist.

Functions

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 19.1

Reverse a function machine



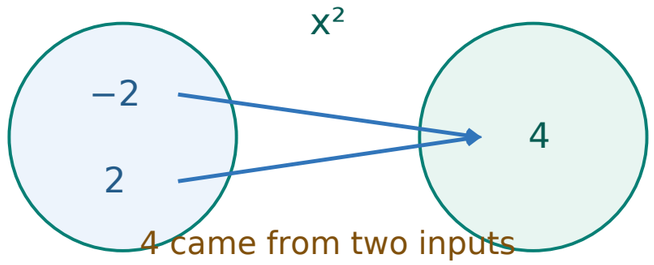
Return: subtract b, then divide by a.

For f(x) = ax + b with a ≠ 0: $f^{-1}(x) = (x - b)/a$.

Watch for: Using the original operation order when reversing.

02 / 19.1

Check whether an inverse is a function

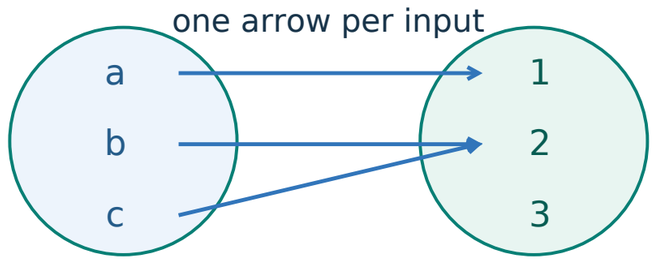


An inverse on the range is a function only when each reached output has a unique original input.

Watch for: Writing ±√x as though it were a single-valued inverse function.

03 / 19.2

Identify domain, range and function validity



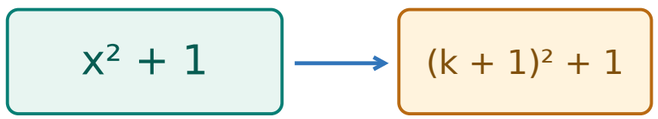
Every x in the domain has exactly one output

range = set of reached outputs ⊆ codomain.

Watch for: Rejecting many-to-one mappings.

04 / 19.3

Substitute an entire input



replace every x by the whole input

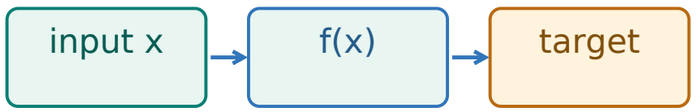
Replace every x by the whole input u: f(u)

In general f(ku) ≠ kf(u).

Watch for: Dropping a negative sign or a cross term from a squared bracket.

05 / 19.3

Find an input for a target output



Solve for the input that reaches the target.

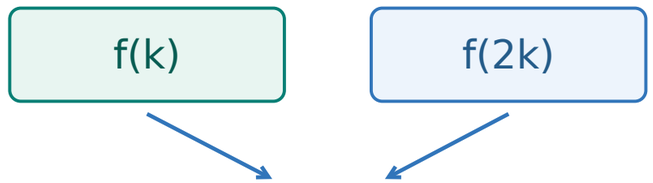
Replace function notation by the corresponding rules

solve in the allowed domain.

Watch for: Reporting the common output when the question asks for the input.

06 / 19.3

Expand separate function values first



Expand each function value separately

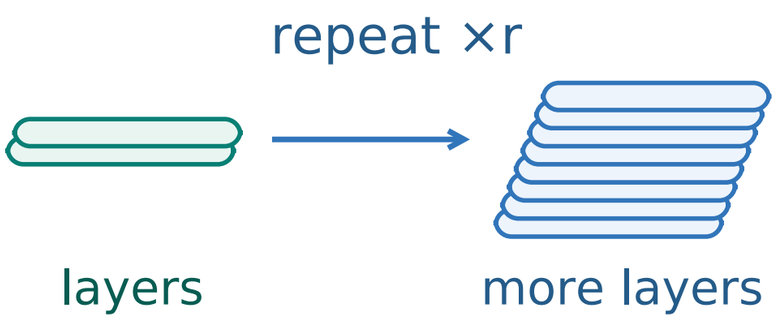
evaluate fixed inputs before solving.

Watch for: Using f(k)+f(2k)=f(3k) without a valid reason.

Functions

07 / 19.3

Model repeated layers or thickness



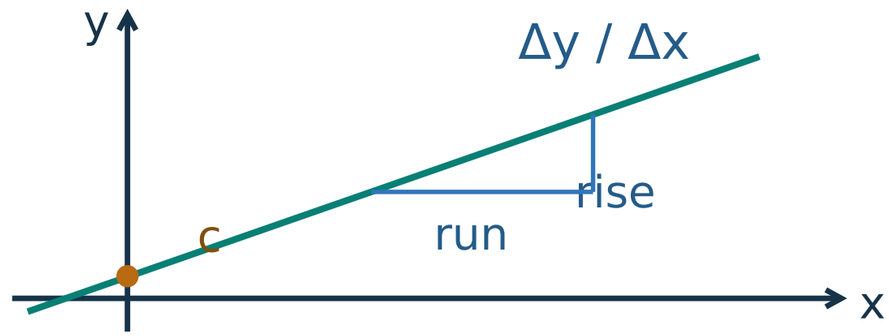
With n operations after the original sheet: $L(n) = r^n$

thickness $H(n) = h_0 r^n$.

Watch for: Using the cut count as the number of layers or mixing millimetres and centimetres.

08 / 19.4

Recover a straight-line function



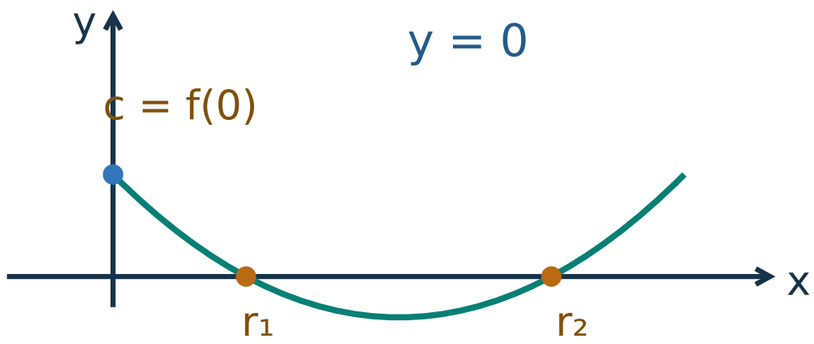
$a = (y_2 - y_1)/(x_2 - x_1)$, $x_2 \neq x_1$

$b = y_1 - ax_1$.

Watch for: Swapping input and output during substitution.

09 / 19.4

Recover a quadratic from graph clues



$f(0) = c$

an x-intercept solves $f(x) = 0$. General a, b, c need three independent conditions unless a coefficient is already known.

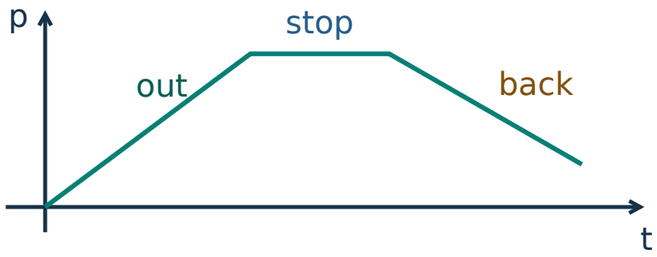
Watch for: Assuming a downward parabola cannot have positive outputs.

Drawing and Interpreting Real-life Graphs

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 20.1

Read movement, stops and return journeys



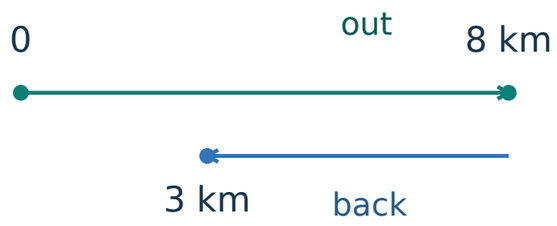
On a segment: signed rate = Δposition/Δtime

speed = |signed rate|.

Watch for: Treating a flat graph at 8 km as movement at 8 km/h.

02 / 20.1

Find total distance and average speed



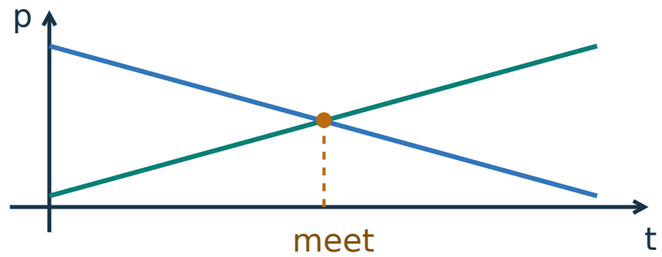
Total distance = Σ|successive position changes|

average speed = total distance/full elapsed time.

Watch for: Calling final position the total distance.

03 / 20.1

Find a meeting or separation



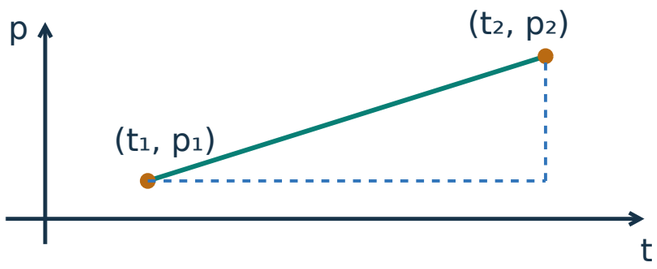
Meeting: pA(t) = pB(t)

separation: |pA(t) − pB(t)| on the same line.

Watch for: Accepting an intersection outside its segment.

04 / 20.1

Write the rule for a graph segment

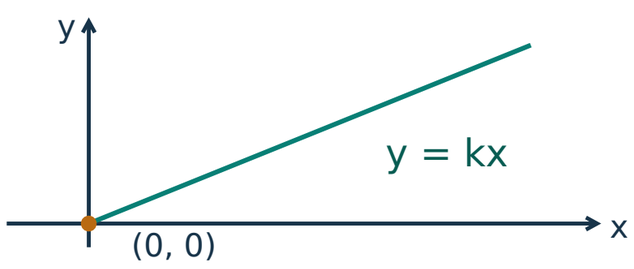


$p(t) = p_1 + [(p_2 - p_1)/(t_2 - t_1)](t - t_1)$, for $t_1 \le t \le t_2$.

Watch for: Assuming every segment begins at the origin.

05 / 20.2

Recognise direct proportion



$y = kx$

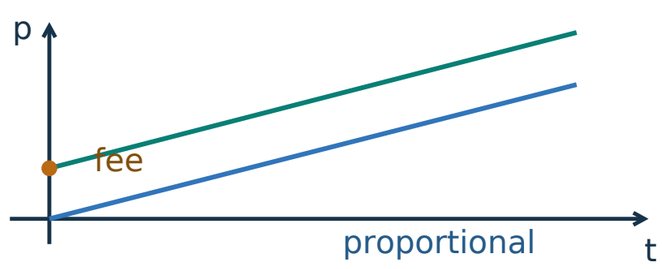
$y/x = k$ for $x \neq 0$

graph passes through (0, 0).

Watch for: Calling y=3.5x+6 directly proportional because it is straight.

06 / 20.2

Separate a fixed fee from a rate



Linear: y = mx + c

directly proportional only when c = 0.

Watch for: Scaling a total price while ignoring its fixed fee.

Drawing and Interpreting Real-life Graphs

07 / 20.3

Predict how fast water height rises



same flow → different height rates

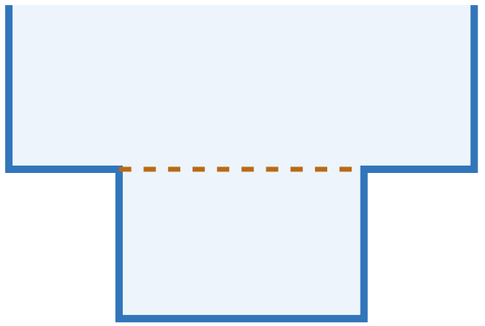
Height rate = Q/A at that level

larger cross-sectional area gives a smaller height rate.

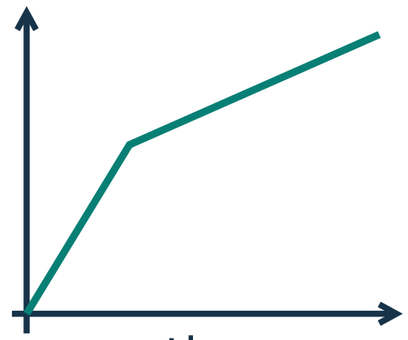
Watch for: Assuming constant flow always gives a straight height-time graph.

08 / 20.3

Join the stages of a filling graph



width changes



time

Time in each part = part volume/Q

cumulative height and time give graph corners.

Watch for: Restarting time when the width changes.

Algebra 2

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 21.1

Add or subtract algebraic fractions



scale each entire numerator

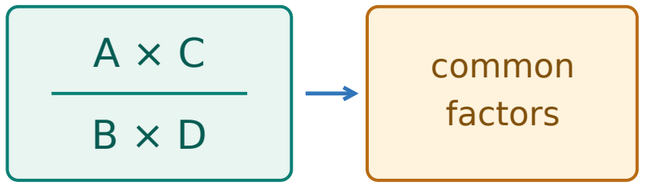
A/B ± C/D = (AD ± BC)/(BD), B, D ≠ 0

use a smaller common denominator when possible.

Watch for: Subtracting only the first term of a bracketed numerator.

02 / 21.1

Multiply fractions by cancelling factors



(A/B)(C/D) = AC/(BD)

only common non-zero factors cancel.

Watch for: Cancelling terms inside a sum.

03 / 21.1

Divide by an algebraic fraction

$$\frac{A}{B} \div \frac{C}{D} \longrightarrow \frac{D}{C}$$

multiply by the reciprocal

(A/B) ÷ (C/D) = (A/B)(D/C), with B, D ≠ 0 and C ≠ 0.

Watch for: Checking denominator restrictions but forgetting division by zero.

04 / 21.1

Substitute a negative or fractional value

use brackets



square the whole signed input

Substitute the whole value in brackets

division by a fraction multiplies by its reciprocal.

Watch for: Treating division by 3/2 as multiplication by 3/2.

05 / 21.2

Clear denominators in an equation



every term on both sides

Multiply both complete sides by the least common denominator

variable factors used must be non-zero.

Watch for: Multiplying only one term on a side.

06 / 21.3

Turn a fraction story into an equation

whole amount X



used part + remaining part

A fraction p of an amount X is pX

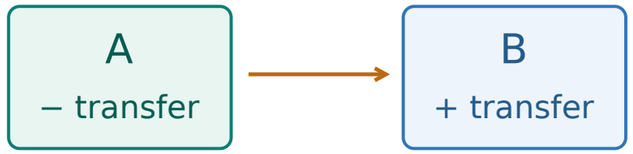
the remainder is (1 − p)X.

Watch for: Applying every fraction to the original amount regardless of the event order.

Algebra 2

07 / 21.3

Track money through successive transfers



total before = total after

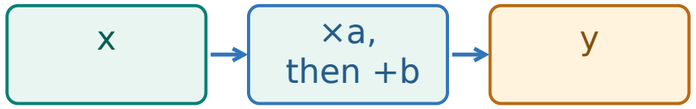
Update both balances after each transfer

total money is conserved if nothing enters or leaves.

Watch for: Taking B’s later transfer as a fraction of the original 30 only.

08 / 21.4

Make a variable the subject



undo: −b, then ÷a

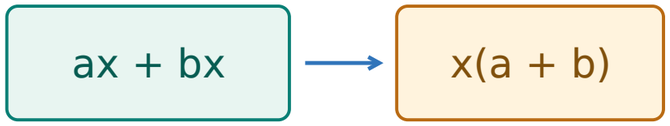
Undo operations in reverse order, keeping equality

division by a coefficient requires it to be non-zero.

Watch for: Changing signs by a slogan without applying the same operation to both sides.

09 / 21.4

Collect a subject that appears twice



one copy of x outside the bracket

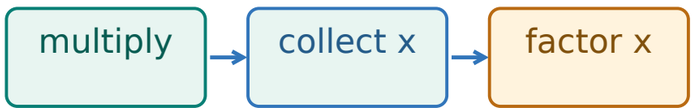
ax + bx = x(a + b)

isolate by factoring, then divide only if the factor is non-zero.

Watch for: Trying to isolate two copies of x independently.

10 / 21.4

Rearrange a fractional formula



retain both denominator restrictions

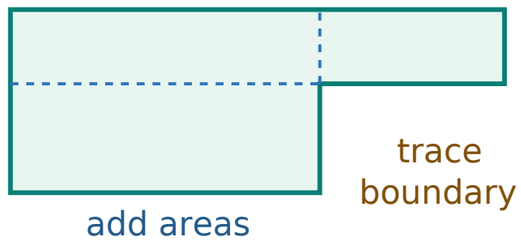
If y = (ax + b)/(cx + d), then x = (b − dy)/(cy − a), provided original cx + d ≠ 0 and cy − a ≠ 0

exceptional cases must be checked.

Watch for: Dividing by an expression that could be zero without checking that case.

11 / 21.5

Build a formula from combined shapes



Perimeter counts the exposed boundary

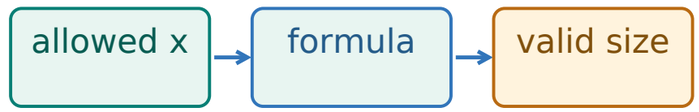
area adds component areas or subtracts removed areas

All physical lengths must be positive.

Watch for: Subtracting the cut-out perimeter from the outer perimeter.

12 / 21.5

Use and reverse a formula within its domain



reject impossible physical answers

Substitution evaluates the rule

reversing solves an equation

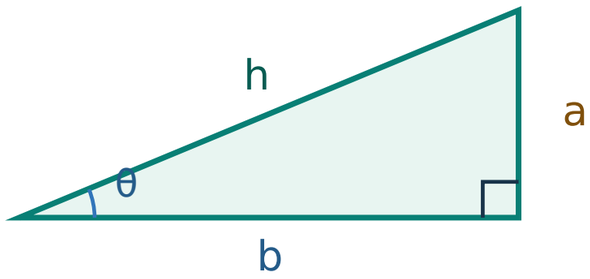
A model’s domain may be narrower than all real numbers.

Watch for: Keeping an algebraic answer that makes a physical length non-positive.

Trigonometry

01 / 22.1

Find a missing side with Pythagoras



$h = \sqrt{(a^2 + b^2)}$

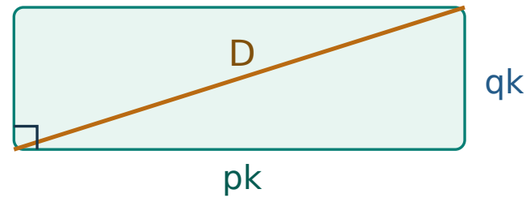
$a = \sqrt{(h^2 - b^2)}$

$h^2 = a^2 + b^2$.

Remember: Right-angled triangle only. Hypotenuse h is longest; h>b>0 when finding a.

02 / 22.1

Use a diagonal and an aspect ratio



width = pk

height = qk

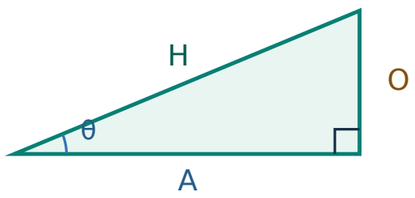
$k = D/\sqrt{(p^2 + q^2)}$

area = $D^2pq/(p^2 + q^2)$.

Watch for: Assuming the longer diagonal always means the larger area for different aspect ratios.

03 / 22.2

Label opposite, adjacent and hypotenuse



$\sin\theta = O/H$

$\cos\theta = A/H$

$\tan\theta = O/A$

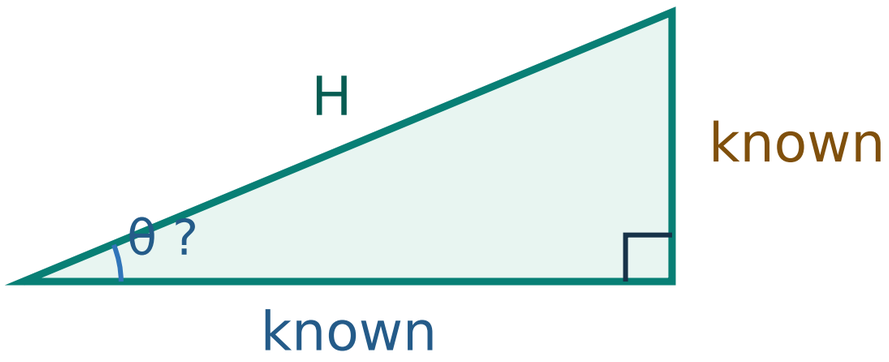
O and A depend on θ

H does not.

Watch for: Calling the bottom side adjacent without looking at the angle.

04 / 22.3

Find an angle from two sides



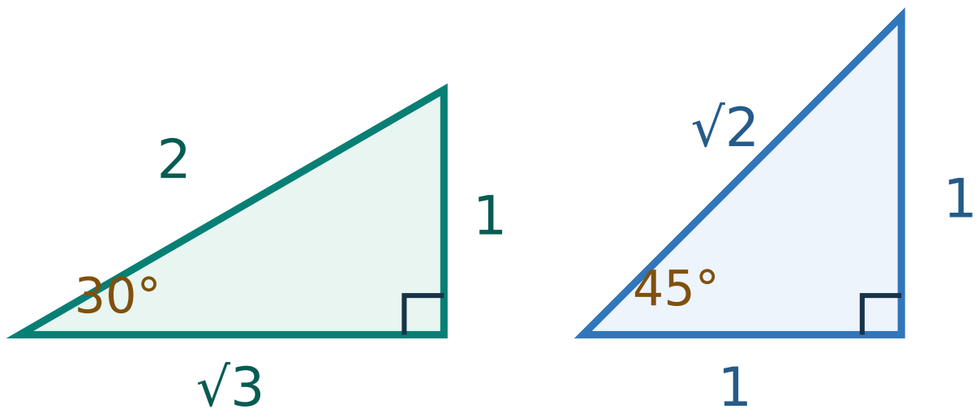
$\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$, in degree mode.

Watch for: Reading $\sin^{-1}$ as 1/sin or using radian mode.

Trigonometry

05 / 22.3

Keep special-angle ratios exact



sin30° = 1/2

cos30° = √3/2

tan30° = √3/3

sin45° = cos45° = √2/2

tan45° = 1

sin60° = √3/2

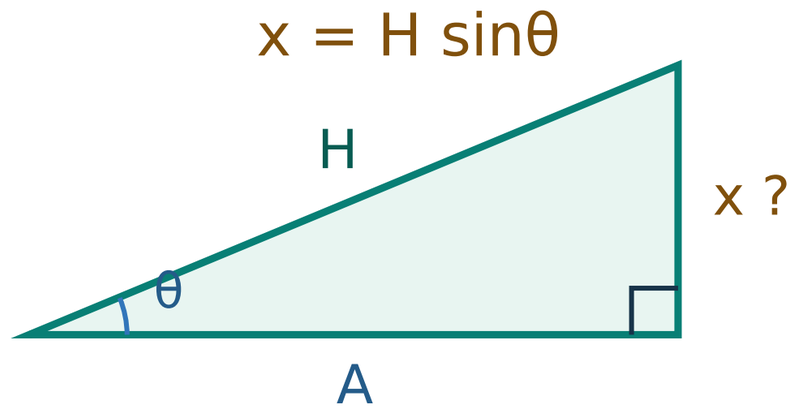
cos60° = 1/2

tan60° = √3.

Watch for: Replacing √3 by 1.7 before later calculations.

06 / 22.4

Find a side above the fraction bar



If sinθ = x/H, then x = H sinθ

similarly for cosine or tangent.

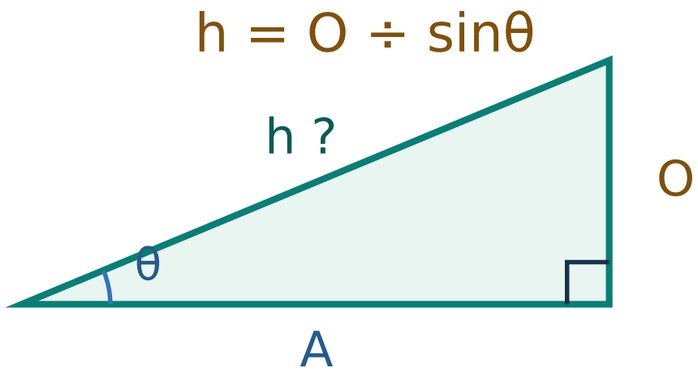
Watch for: Choosing multiplication from memory without first writing the equation.

Trigonometry

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

07 / 22.4

Find a side below the fraction bar



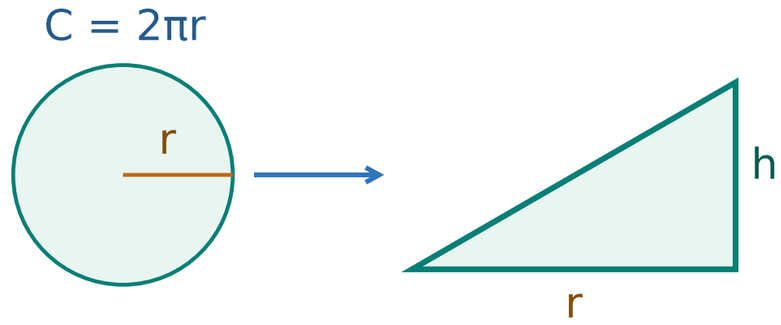
If sinθ = O/h, then h = O/sinθ

if tanθ = O/a, then a = O/tanθ.

Watch for: Multiplying 7 by sin30° when the unknown is underneath.

08 / 22.5

Convert a curved measurement to a distance



C = 2πr

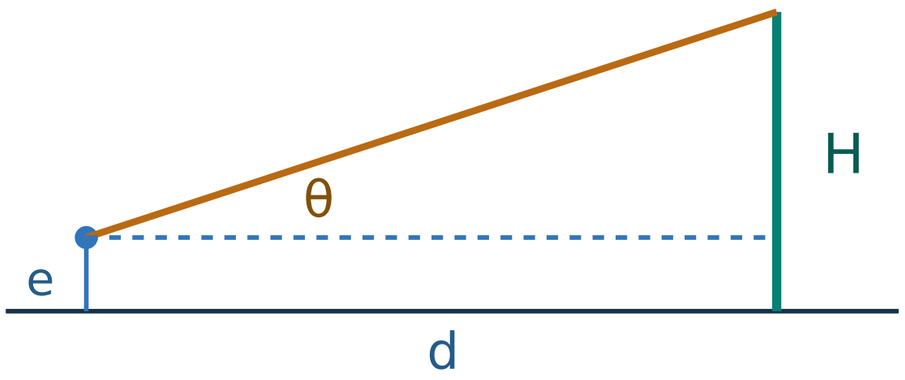
r = C/(2π)

The right triangle needs horizontal distance, not curved boundary length.

Watch for: Using the circumference as the triangle’s horizontal leg.

09 / 22.5

Add eye height to the calculated rise



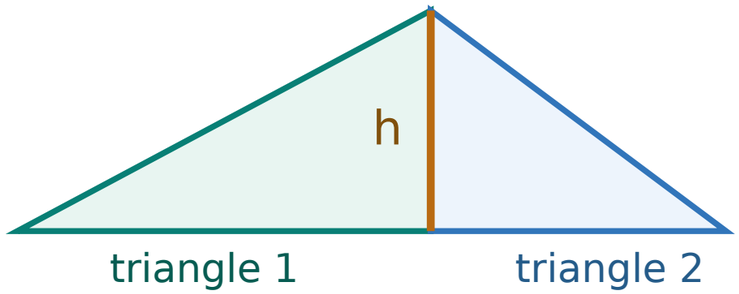
H = e + d tanθ when e is eye height and d is horizontal distance

adjust the vertical reference if ground levels differ.

Watch for: Adding eye height twice or confusing a sloping sight line with d.

10 / 22.5

Link triangles through a shared side



Carry the shared side exactly

$(a\sqrt{b})^2 = a^2b$

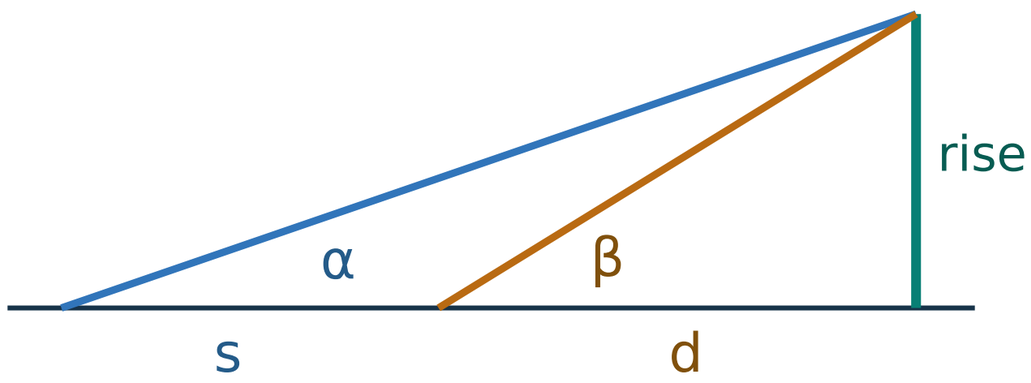
Each triangle needs its own right angle and reference angle.

Watch for: Adding angles without checking the rays’ order.

Trigonometry

11 / 22.5

Link two observation positions



Same vertical rise gives d tanβ = (d + s)tanα when the farther observer is s farther away, at the same eye level.

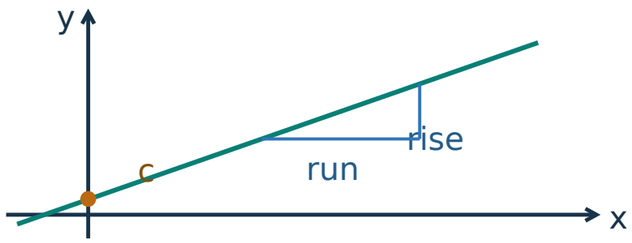
Watch for: Treating the distance between observers as either complete triangle base.

Graphing Functions

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 23.1

Build a straight-line graph



$m = \Delta y/\Delta x$ for $\Delta x \neq 0$

$c = f(0)$

$y - y_1 = m(x - x_1)$.

Watch for: Plotting the constant c on the x-axis.

02 / 23.1

Recover a linear rule from a table

x	y
2	11
5	20

→ $\Delta y = 9$, $\Delta x = 3$, $m = 3$

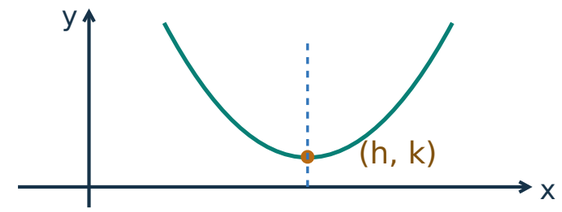
m = (output change)/(input change)

fixed part $c = y - mx$.

Watch for: Dividing total cost by hours when there is a fixed fee.

03 / 23.2

Find and plot a quadratic's turning point



$y = ax^2 + bx + c$, $a \neq 0$

vertex $x = -b/(2a)$

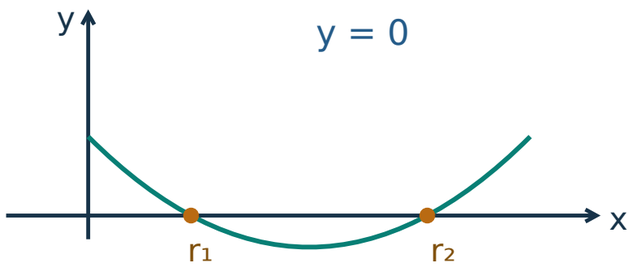
$y = f(-b/(2a))$

$y = a(x - h)^2 + k$ has vertex (h, k).

Watch for: Drawing a quadratic as a broken straight-line graph.

04 / 23.2

Find roots and axis intercepts



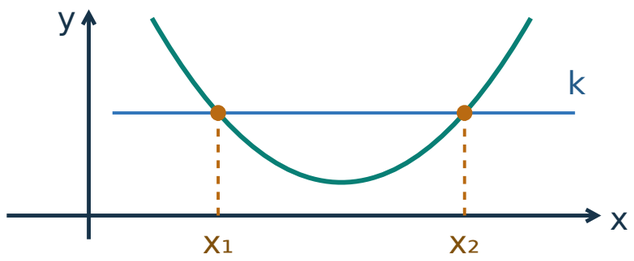
y-intercept: x = 0

x-intercepts: f(x) = 0. For a quadratic, factor when possible and use the zero-product rule.

Watch for: Calling a root (0,2) instead of (2,0).

05 / 23.3

Read inputs for a target output



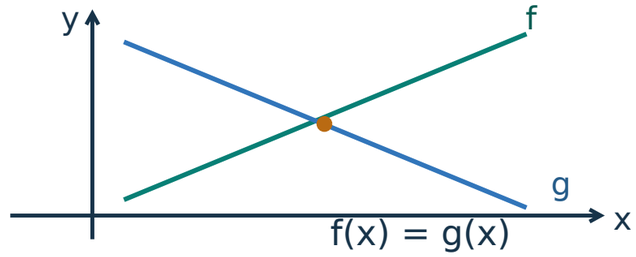
Draw the horizontal line y = k

read all intersection x-values.

Watch for: Drawing x=k or missing the second crossing.

06 / 23.3

Find where two functions agree



f(x) = g(x) at their intersections

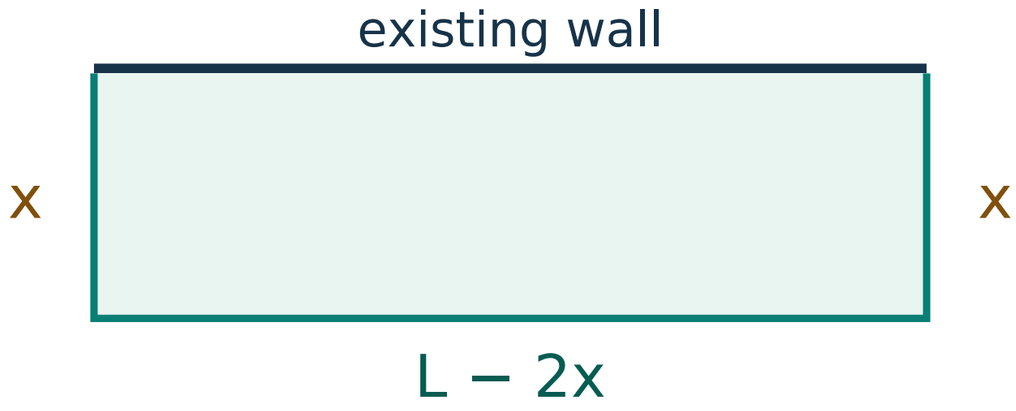
equal coordinates use the same x and y.

Watch for: Confusing an intersection of two functions with an x-axis root.

Graphing Functions

07 / 23.4

Build an area model with three fenced sides



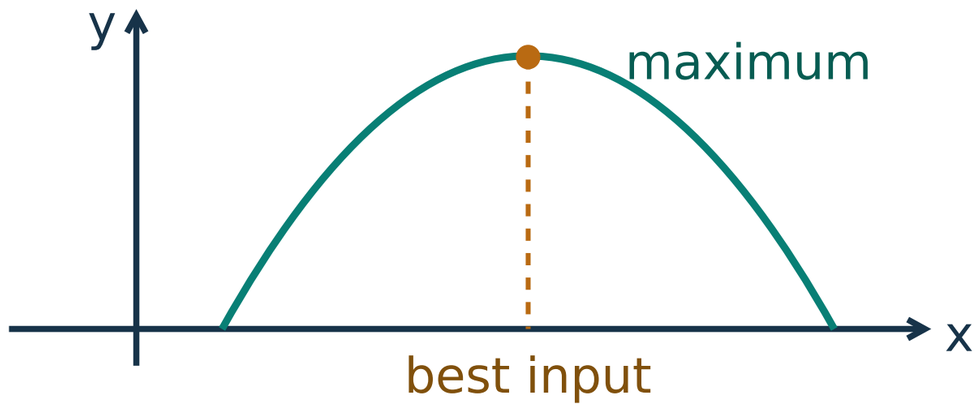
For three fenced sides with total L: length = L − 2x

A = x(L − 2x), with 0 < x < L/2.

Watch for: Using a four-sided perimeter when one side is an existing wall.

08 / 23.4

Find and interpret a maximum or minimum



For $A = ax^2 + bx + c$, vertex input = $-b/(2a)$

check domain endpoints when included

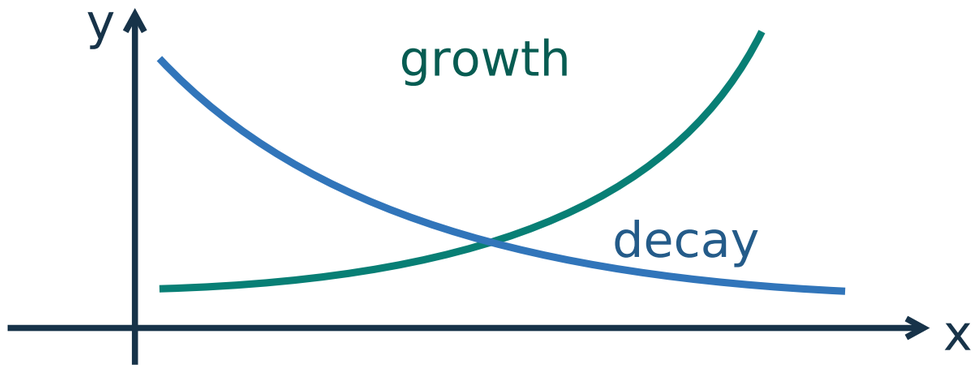
Three-side fencing: maximum $L^2/8$ at x = L/4, length = L/2.

Watch for: Giving the maximum’s input instead of the maximum output.

Graphing Functions

09 / 23.5

Recognise exponential growth or decay



$y = ab^x$ with $a > 0$, $b > 0$, $b \neq 1$

$y(0) = a$

$y(x + 1)/y(x) = b$

$b > 1$ gives growth

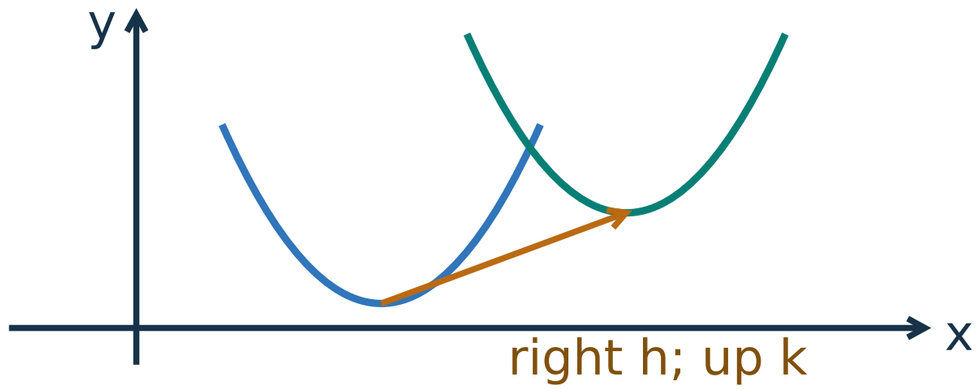
$0 < b < 1$ gives decay

$b^{-n} = 1/b^n$.

Watch for: Expecting equal additive gaps or treating a graph estimate as exact.

10 / 23.5

Transform a familiar graph



$f(x) + k$: up k

$f(x - h)$: right h

$-f(x)$: reflect in x-axis

$f(-x)$: reflect in y-axis

Combined $g(x) = a\,f(x - h) + k$ maps (u, v) to $(u + h, av + k)$.

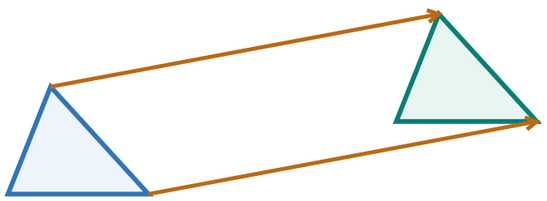
Watch for: Reading $x-2$ as a shift left instead of right.

Geometry 3: Transformations - Constructions

FIND YOUR TASK → MATCH THE PICTURE → CHECK THE CONDITION → USE THE FORMULA

01 / 24.1

Translate every vertex by the same vector



same vector for every vertex

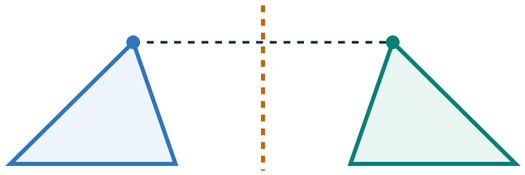
Vector (a, b): (x, y)↦(x + a, y + b)

image − original = (a, b) for every vertex.

Watch for: Finding a vector from one vertex but using a different one for another.

02 / 24.2

Reflect in the stated mirror line



mirror line

In x = a: (x, y)↦(2a − x, y)

in y = b: (x, y)↦(x, 2b − y)

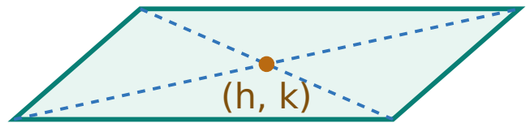
in y = x: (y, x)

in y = −x: (−y, −x).

Watch for: Accepting a mirror that works for one pair but fails for the remaining vertices.

03 / 24.2

Distinguish central and axial symmetry



half-turn about the centre

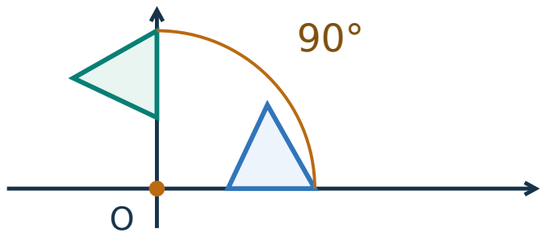
Central symmetry about (h, k): (x, y)↦(2h − x, 2k − y)

Axial symmetry means reflection in a line maps the entire shape to itself.

Watch for: Treating central symmetry as reflection in an arbitrary line through the centre.

04 / 24.3

Rotate about the correct centre



About origin: 90° anticlockwise (−y, x), 90° clockwise (y, −x), 180° (−x, −y)

About (h, k), translate to the origin, rotate, then translate back.

Watch for: Giving only “a rotation” or rotating around the wrong centre.

05 / 24.3

Describe a complete transformation



Translation needs a vector

reflection needs a mirror line

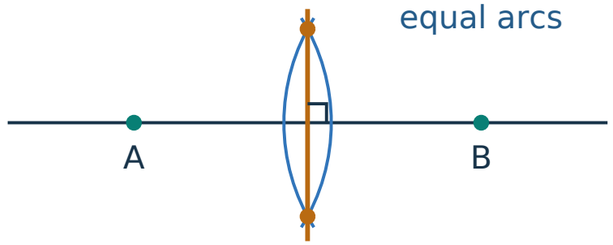
rotation needs centre, angle and direction

An ordered mapping fixes the vertex correspondence.

Watch for: Calling a point match enough evidence for a whole-shape mapping.

06 / 24.4

Construct a perpendicular bisector



Use equal radius r > AB/2 from A and B

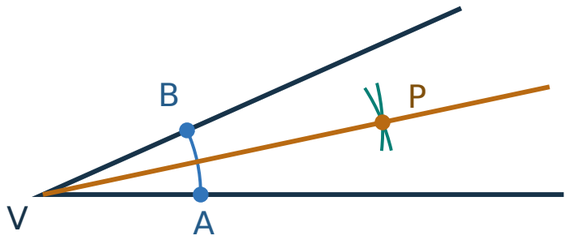
joining the two intersections gives the perpendicular bisector.

Watch for: Changing the compass opening between A and B.

Geometry 3: Transformations - Constructions

07 / 24.4

Bisect an angle with equal arcs



From vertex V mark VA = VB

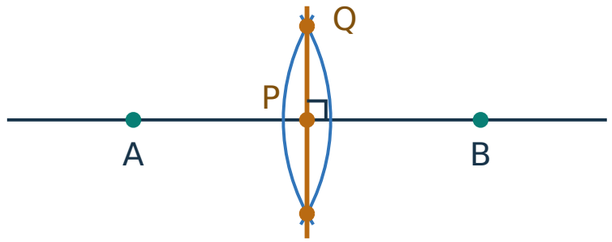
find P with PA = PB using equal arcs

ray VP bisects ∠AVB.

Watch for: Joining A to B instead of V to P.

08 / 24.4

Construct a perpendicular on a line



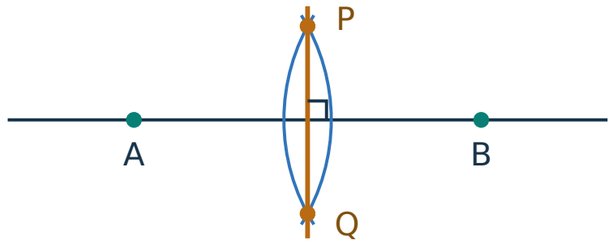
Mark PA = PB on opposite sides

a point Q with QA = QB gives PQ perpendicular to AB.

Watch for: Placing A and B at unequal distances from P.

09 / 24.4

Construct a perpendicular from outside



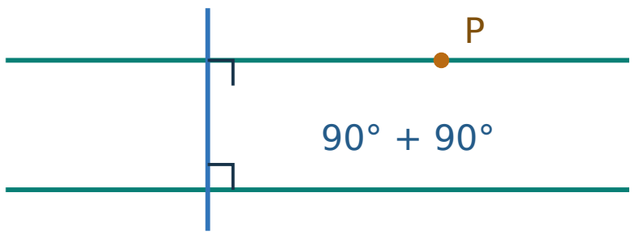
One circle centred at P cuts the line at A, B with PA = PB

a second equal-distance point Q fixes their bisector PQ.

Watch for: Drawing from P to just one crossing A instead of the matching-arc intersection.

10 / 24.4

Construct a parallel through a point

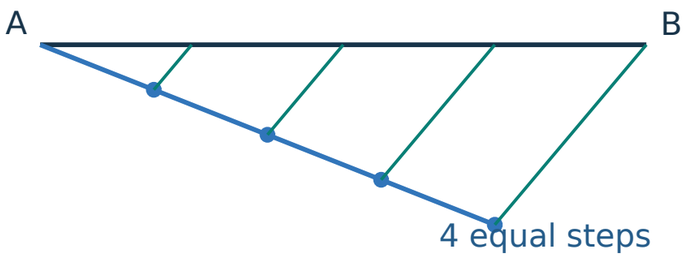


Two distinct coplanar lines perpendicular to the same line are parallel.

Watch for: Drawing a parallel-looking line that misses the specified point.

11 / 24.4

Divide a segment into equal parts



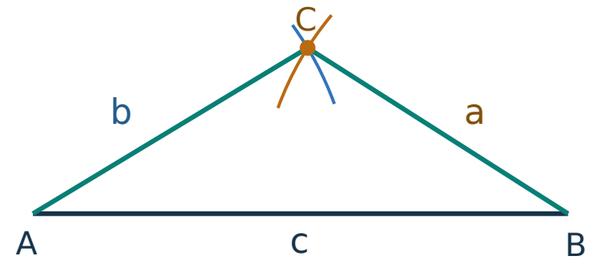
Equal steps on an auxiliary ray transfer to equal parts of AB through parallel lines

n is a positive integer.

Watch for: Using n−1 auxiliary steps for n required parts.

12 / 24.5

Construct a triangle from three sides



Each side must be positive

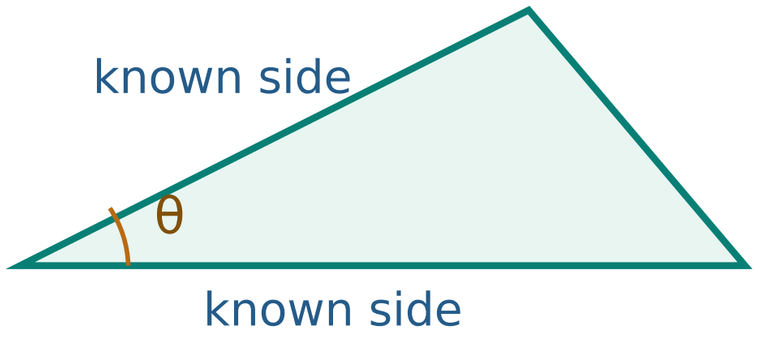
the two shorter lengths must sum to more than the longest.

Watch for: Using one radius for both arcs when AC and BC differ.

Geometry 3: Transformations - Constructions

13 / 24.5

Construct a triangle from side-angle-side



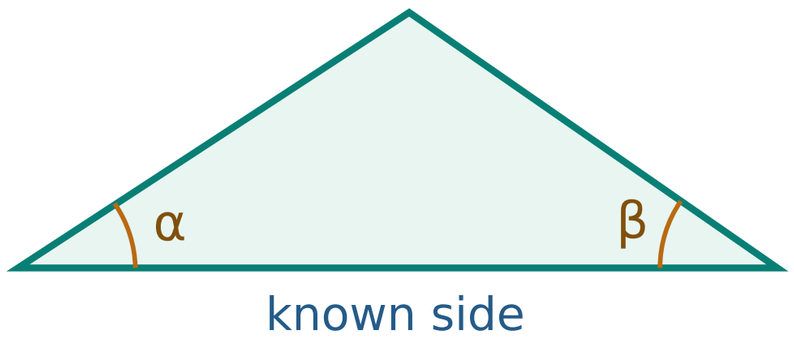
The angle lies between the two known sides

lengths > 0 and 0° < included angle < 180°.

Watch for: Placing the supplied angle opposite one given side instead of between them.

14 / 24.5

Construct a triangle from angle-side-angle



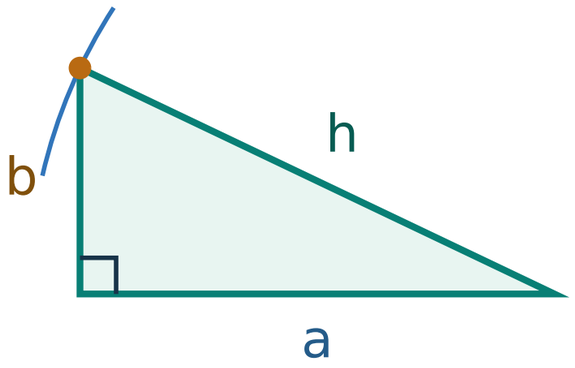
For endpoint angles α, β > 0, require α + β < 180°

the third angle is 180° − α − β.

Watch for: Measuring B’s angle from the outward continuation rather than BA.

15 / 24.5

Construct a right triangle from RHS



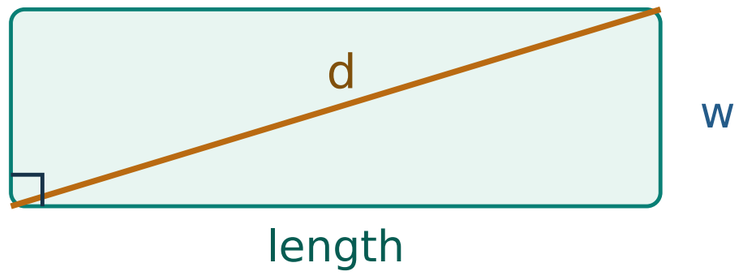
Hypotenuse h must exceed the given leg a > 0

the other leg is $\sqrt{(h^2 - a^2)}$.

Watch for: Centring the hypotenuse arc at the right-angle vertex.

16 / 24.5

Construct and check a rectangle



Opposite sides equal and parallel

all angles 90°

diagonal² = length² + width².

Watch for: Using the diagonal as the rectangle’s width.